

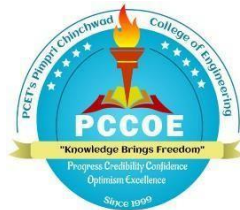
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A PROJECT REPORT ON
APP-BASED AUDIOMETER FOR HEARING LOSS
DETECTION AND PREDICTION USING MACHINE
LEARNING

SUBMITTED TO THE PIMPRI CHINCHWAD COLLEGE OF
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REQUIREMENTS FOR THE AWARD
OF THE DEGREE
OF

BACHELOR OF TECHNOLOGY
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SUBMITTED BY

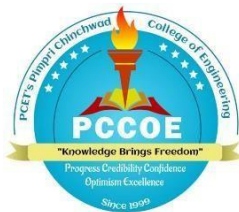
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CERTIFICATE

This is to certify that the project report entitles
**“APP-BASED AUDIOMETER FOR HEARING LOSS DETECTION AND
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ABSTRACT

Hearing loss is a growing public health concern that affects people of all ages, often going unnoticed until it reaches an advanced stage. One of the key reasons for delayed diagnosis is the lack of accessible and affordable hearing assessment tools. Conventional audiometers, though accurate, are costly, require trained personnel, and are not easily portable. These limitations make them unsuitable for widespread screening, particularly in rural or underserved regions where healthcare infrastructure is limited.

This research project addresses this challenge by designing and developing a mobile-based digital audiometer powered by machine learning techniques. The core idea is to create a portable, cost-effective, and user-friendly solution for early detection of hearing loss. The application collects basic user information, such as age, along with audiometric data from both ears. This data is then processed using a trained Random Forest algorithm, which analyzes hearing patterns and predicts the presence of hearing impairment.

The app not only provides a digital platform for hearing tests but also enables left vs. right ear comparison, stores historical data for long-term tracking, and presents the results in a clear and accessible format. The use of the Random Forest algorithm was found to deliver high accuracy in predictions, making it a reliable tool for preliminary screening.

By combining healthcare, mobile technology, and machine learning, this project offers a scalable solution that can be deployed in schools, primary health centers, and community outreach programs. It empowers users to take early action and seek medical advice before the condition worsens. The system can also serve as a support tool for ENT professionals by providing pre-assessment data.

In conclusion, this research demonstrates that a digital, app-based audiometer using machine learning can effectively overcome the limitations of traditional systems. It holds significant potential to make hearing care more accessible, proactive, and inclusive—especially for populations with limited access to clinical facilities.

KEYWORDS - Hearing Loss(HL), Random Forest Classifier(RFC), Audiometer, Calibration of Headset.

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LIST OF ABBREVIATIONS

Abbreviation	Illustration
HL	Hearing Loss
RFC	Random Forest Classifier
SVM	Support Vector Machine
MLP	Multi-layer Perceptron
ABA	App-based Audiometer
ENT	Ear-Nose-Throat

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Chapter 1

Introduction

Hearing loss is a significant public health concern, with projections from the World Health Organization (WHO) indicating that over 430 million people will experience disabling hearing loss by 2030, a number expected to rise to 700 million by 2050. Early detection and intervention are crucial in mitigating the adverse effects of hearing impairment, which extend beyond auditory function to impact cognitive development, social interactions, educational achievements, and overall quality of life. Conventional hearing assessments rely on pure-tone audiometry (PTA), which, while highly accurate, requires specialized equipment and trained audiologists. The cost of traditional audiometers ranges from several thousand to tens of thousands of dollars, making them inaccessible in low-resource settings. In rural areas, individuals often travel long distances to reach specialized clinics, delaying necessary interventions. The widespread availability of mobile devices presents an opportunity to democratize hearing healthcare. However, ensuring accuracy in these tests is challenging due to device variability and environmental factors such as background noise. Developing reliable mobile audiometry solutions requires careful consideration of hardware limitations, calibration, and user interface design to maintain testing validity.

Furthermore, traditional methods are often inefficient when mass testing is required (e.g., schools or clinics). The prevalence of hearing impairment determined the need for testing, the proportion of normal-hearing patients undergoing testing, and the assumed testing frequency. Capacity is assessed based on the number of audiologists and the typical daily audiograms conducted. Time savings were estimated using the average audiogram duration, with an assumption that 80% of the process could be automated. While essential for early intervention, school hearing screening programs can be logistically complex and time-consuming when using conventional approaches. To address these challenges, this research proposes the development of an App-Based Audiometer (ABA) for hearing loss detection and prediction using ML. The Hearing Test application for Android was launched in 2013 and has been consistently improved ever since. These applications are specifically designed to be utilized within an Individual Testing Model, allowing for comprehensive personal assessments, while also possessing the capability to be effectively employed for large-scale mass screening to facilitate broader population-level evaluations.

In recent years, ML has emerged as a powerful tool in healthcare, offering new possibilities for predicting hearing loss. These advanced algorithms can identify individuals at risk of hearing impairment by analyzing large datasets that include audiometric tests, medical histories, and demographic information. The use of ML in hearing healthcare represents a paradigm shift from reactive to proactive intervention strategies. ML algorithms are employed to analyze audiometric data, improving diagnostic precision and enabling personalized, proactive hearing assessments. The potential impact of this technology extends beyond individual health outcomes to include broader societal benefits, such as reduced healthcare costs, improved educational outcomes, and enhanced workplace productivity. Through a combination of mobile technology, ML algorithms like Xgboost, SVM, RFC, Naive Bayes, and Adaboost, and user-centered design, this research contributes to the ongoing transformation of hearing healthcare accessibility and delivery.

1.1 Overview

The project titled "App-Based Audiometer for Hearing Loss Detection and Prediction Using Machine Learning" falls under the domains of Digital Healthcare, Mobile Computing, and Machine Learning. This innovative solution was designed and developed to address the increasing prevalence of hearing loss globally, particularly in rural and resource-constrained environments where traditional audiometric equipment is either unavailable or prohibitively expensive.

According to the World Health Organization (WHO), over 700 million people are projected to suffer from disabling hearing loss by 2050. Traditional Pure Tone Audiometry (PTA) systems often cost thousands of rupees and require specialized setups, making them inaccessible for the general population in underserved regions. Our project offers a cost-effective, portable, and intelligent mobile application that can perform accurate hearing assessments using machine learning models integrated into a Flutter-based mobile app with a Firebase backend.

The application not only conducts hearing tests by generating standardized tones but also uses noise calibration techniques to ensure accurate results even in noisy environments (≤ 40 dB). The system predicts hearing loss severity — categorized as Normal, Moderate, or Severe — using trained machine learning models such as Support Vector Machines (SVM - 96% accuracy), Random Forest Classifiers, and XG-Boost.

This project gained national-level recognition as a Finalist at Smart India Hackathon 2023 (SIH '23) and emerged as the Winner of the IDE Boot Camp 2024, validating its real-world impact and technical rigor. The solution demonstrated 80% faster screening times compared to traditional methods and achieved 96% agreement with clinical PTA tests across 25 participants, supported by a dataset of 1,000+ audiograms.

The app has already been piloted for mass screening programs in schools and workplaces, confirming its scalability and usability. Furthermore, it has the potential to integrate with telemedicine platforms and smart hearing aids in the future, extending its benefits to continuous remote monitoring and diagnostics.

By combining accessible mobile technology with intelligent healthcare diagnostics, this project aims to make early detection of hearing loss both affordable and widespread, ultimately contributing to improved quality of life and healthcare inclusivity.

1.2 Motivation

- **Rising Hearing Issues** – Increasing cases of hearing impairments.
- **Early Detection & Prevention** – Timely identification of hearing issues can prevent further deterioration.
- **Affordability & Scalability** – A digital, low-cost alternative to conventional audiometers, making hearing tests accessible to a larger population.
- **ML-powered Analysis** – Enhancing accuracy and enabling proactive healthcare decisions through machine learning.

1.3 Problem Statement and Objectives

Problem Statement:

App-Based Audiometer for Hearing Loss Detection and Prediction Using Machine Learning

Hearing loss affects millions worldwide, yet traditional audiometry tests are costly, time-consuming, and inaccessible in many areas. A portable, user-friendly, and cost-effective solution for hearing assessment is needed.

Our project introduces an App-Based Audiometer that conducts hearing tests. The audiograms collected are loaded in an ML classifier, Random Forest, which predicts hearing loss severity.

Objectives:

- 1) To design and develop a Mobile Application for conducting hearing tests using smartphone hardware, eliminating the need for traditional audiometry equipment.
- 2) To evaluate the accuracy, sensitivity, and specificity of the developed solution, ensuring its reliability as an alternative to traditional audiometer.
- 3) To collect, integrate, and analyze hearing test data for generating data-driven insights and tracking hearing health over time.
- 4) To develop a Machine learning model for analyzing and comparing hearing issues in the left and right ear, identifying patterns and trends to enable better healthcare interventions.

1.4 Scope of the work

- **Scope:**

1. Smartphone-Based Hearing Tests – Enables hearing assessments without the need for specialized hardware.
2. Cost-Effective & Accessible – Provides an affordable solution, particularly for rural and underserved areas.
3. User Data Management – Stores test results securely for future analysis and tracking.
4. Early Detection & Intervention – Helps identify hearing impairments at an initial stage to prevent worsening conditions.

Chapter 2

Literature Survey

2.1 Literature Review

ABA has been an active area of research over the past two decades, focusing on automated audiogram acquisition and interpretation through mobile and web-based solutions. The background noise level during the ABA test should not exceed 35–40 dB, which was monitored using the built-in microphone of the mobile device , to ensure accurate evaluation of hearing thresholds as low as 0 dB HL. Likewise, mobile audiometry should be conducted under similar conditions for reliable results. The development of an ABA requires careful calibration . Two critical aspects of this process are device calibration and active noise monitoring/calibration. Several smartphone-based digital audiometer tools have been developed and marketed to bridge the gap between the availability of hearing professionals and the increasing demand for accessible hearing assessments . Several tablet-based audiometers have been developed to enable hearing assessments beyond traditional soundproof booths. These mobile solutions are now utilized globally, providing essential hearing evaluations in remote areas with limited access to hearing professionals . Proper calibration ensures accurate and reliable hearing assessments.

This helps maintain consistency across different devices and environments. To conduct a hearing test using an ABA, the application provided an instruction for conducting hearing tests, the hearing tests were performed according to the app's algorithm at the frequencies of 250 Hz, 500 Hz, 1000 Hz, 2000 Hz, 4000 Hz, and 8000 Hz. For Ex: The test typically begins with a 1000 Hz signal at 40 dB, presented through headphones to the subject. If the tone is heard, the subject responds by selecting the "Yes" option. If the tone is not perceived, they must select "No" instead. The test tone's intensity then increases in 10 dB increments until the subject detects it . Then these values are stored in the database to generate the Audiogram. This stored Audiogram data is used for further proposed ML Models.

Artificial Intelligence and ML have been used in medical research, including audiology, radiology, dermatology, cancer, and ophthalmology . Several audiogram interpretation models have been developed to address the challenge of automatically extracting such information from audiograms.

Concerning audiometric data specifically, one group has deployed ML algorithms to automatically estimate audiogram performance based on online participant testing. Four machine learning algorithms (RF, Adaboost, MLP, SVM) were developed to predict noise-induced hearing loss. However, no prior models have been developed to interpret existing traditional plot-based audiogram data automatically. The purpose of this manuscript is to present a novel ABA along with an ML model that accurately and quickly interprets the accuracy of hearing loss separately for the left and right ear from audiograms.

2.2 Gap Identification / Common findings from the literature:

Table 2.1 Comparison table of Gap Identification

Ref.	Dataset	Tech/Algo.	Performance	Challenge/Limits/ Future Scope
[4]	70 audiograms	Analysis – Pearson’s correlation (accuracy), Bland-Altman (agreement & bias), classification accuracy (detection).	Class A(<35dB):97.4% Class B(35-64dB):22.2% Class C(>65dB):38.5%	Inconsistent across devices. Severe Cases – Lower detection accuracy. Small Sample.
[9]	256 audiograms	Calibration: External device connected to headphones to calibrate it. Test: Taken by minimum threshold of intensity at a frequency that is heard.	Sensitivity: 93.8% Specificity:82.2%	1)Dataset Limitations – The study used only 256 audiograms, restricting accuracy. 2)Limited Classification Scope
[10]	250 audiograms	Took Test on Hearing Test app PTA Testing Result Comparison	Sensitivity 29% Specificity 100%	1) Calibration issues arise as earphones aren't directly calibrated. 2) Constant display of previous results may bias retests, reducing score differences.
[11]	Own dataset	Gradient Boosting classifier (GBC),Random Forest Classifier (RFC)	Test accuracy: 92.6% Train accuracy:93.625%	Improving accuracy and expanding influence of these models.

[20]	1,644 patients	Random Forest Classifier(RFC), Adaboost, MLP,SVM	RF:-79.7% Adaboost:-78.6% MLP:-78.9% SVM:-80.1%	Future model refinements and larger data sets are expected to increase accuracy
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Chapter 3

Software Requirement Specification

3.1 Functional Requirements

3.1.1 System Features (Functional Requirements)

1. **User Authentication:** Ensures a secure and seamless login process, provides easy access, and safeguards user data with robust protection.
2. **Data Storage & Analysis:** Efficiently storing, organizing, and processing hearing test data for meaningful insights.
3. **Background Noise Calibration:** Measure ambient noise; proceed only if ≤ 40 dB. Alert users if the noise is too high and provide a retry option.
4. **Headset Calibration:** Ensure headphones are connected. Play **80 dB tone**; users adjust the volume until consistently heard. Confirm calibration before proceeding.
5. **Hearing Test Execution:** Generating highly accurate and well-calibrated audio tones while meticulously capturing and analyzing user responses with precision.
6. **Report Generation:** Delivering comprehensive reports that provide users with valuable hearing health evaluations.

3.2 External Interface Requirements

3.2.1 User Interfaces

1. Provides a structured and user-friendly interface for:
 - a) Logging in and accessing hearing tests.
 - b) Headset calibration and noise level checks.
 - c) Viewing test results in an audiogram format.
2. Displays clear instructions and visual indicators for guidance.

3.2.2 Hardware Interfaces

1. Supports both wired and Bluetooth headphones for hearing tests.
2. Uses the device microphone to measure ambient noise before testing.

3.2.3 Software Interfaces

1. Integrates Firebase for:
 - a) User authentication.
 - b) Storing and retrieving test reports.
2. Developed using Flutter, ensuring compatibility with Android and iOS.

3.3 Nonfunctional Requirements

3.3.1 Performance Requirements

- 1 **Performance:** The application delivers a seamless and responsive user experience across a wide range of Android and iOS devices. It ensures minimal latency in tone generation, real-time hearing test execution, and result computation. The system is optimized for efficient memory and CPU usage, thereby preventing excessive battery consumption, overheating, or lag—even on low-end smartphones. Extensive testing was carried out to validate performance consistency under various environmental conditions, including rural field settings.
- 2 **Usability:** The app features a clean, user-friendly interface designed with accessibility and inclusivity in mind. It follows human-centered design principles to ensure that users with no technical background can perform hearing tests effortlessly. Step-by-step guidance, visual cues, and feedback messages help users navigate through tone calibration, test initiation, and result interpretation. The layout is responsive, with large buttons and readable fonts, suitable for both elderly and younger populations.
- 3 **Security:** To maintain the confidentiality and integrity of user health data, the app integrates Firebase Authentication for secure login and role-based access control. All user data—including test results, profiles, and session logs—is encrypted during transmission and securely stored in the Firebase Realtime Database. The application complies with standard data protection guidelines and ensures that no personally identifiable information (PII) is shared or exposed to unauthorized entities.
- 4 **Scalability:** The application architecture is designed to support high user concurrency and future growth. It can accommodate a large number of simultaneous users without performance degradation. The backend infrastructure allows easy scaling of storage and compute capacity, facilitating the integration of advanced features such as real-time analytics, AI-driven voice monitoring, and cloud-based patient history. The modular design ensures that upcoming enhancements (e.g., multi-language support, smart hearing aid connectivity, telehealth modules) can be implemented without major code refactoring.

- 5 **Reliability and Fault Tolerance:** The system is built with fault tolerance in mind. In case of unstable internet connections or device disruptions, the app ensures safe recovery and resumes test progress where left off. Offline capabilities are also supported, allowing data to be temporarily cached and synced once connectivity is restored.
- 6 **Maintainability:** The codebase follows best practices such as modular design, code documentation, and use of version control (Git), ensuring ease of maintenance and collaboration. This design supports iterative improvements and long-term sustainability of the application.

3.3.2 Safety / Security Requirements

- 1 **User Authentication & Access Control** – The application implements secure login mechanisms using Firebase authentication to prevent unauthorized access. Only registered users can access hearing test data.
- 2 **Data Encryption** – All user data, including test results and personal information, is encrypted during transmission and storage to ensure confidentiality and prevent data breaches.
- 3 **Privacy Compliance** – The app adheres to standard data privacy policies, ensuring that no user information is shared with third parties without consent.
- 4 **Secure Data Storage** – Test results and user details are stored on a secure firebase database with restricted access, preventing unauthorized modifications or leaks.
- 5 **Error Handling & Fail-Safe Mechanisms** – The app includes safeguards to prevent incorrect test results due to sudden app crashes or hardware failures, ensuring reliable hearing assessments.
- 6 **Regular Security Updates** – The application framework and dependencies are periodically updated to address security vulnerabilities and maintain data protection.

3.4 System Requirements:

3.4.1 Database Requirements

- 1 The project leverages **Firebase Realtime Database** as the primary backend solution for secure and efficient data storage. It plays a crucial role in maintaining the integrity, accessibility, and real-time availability of user data, hearing test results, and audiometric patterns.
- 2 **Real-Time Synchronization:**
One of the most significant features of Firebase is its real-time data synchronization

capability. As soon as a user completes a hearing test, the results are instantly uploaded and synced across all devices linked to the same account. This ensures that healthcare professionals, users, and administrators can access the latest test records without any delays, making the application suitable for remote diagnosis and continuous monitoring.

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3 **Structured and Hierarchical Data Storage:**

Firebase organizes data in a structured, tree-like JSON format, which facilitates easy retrieval, update, and deletion operations. Hearing test records are stored with relevant metadata such as timestamp, ear tested (left/right), decibel levels, frequency responses, and predicted severity class. This structure supports efficient querying and filtering of data for individual users, groups, or batch screenings (e.g., school-wide campaigns).

4 **Data Security and Privacy Compliance:**

All data transactions are encrypted both in transit and at rest. Firebase security rules are configured to control read/write access based on authenticated user roles, ensuring that sensitive medical information remains private and protected. The app follows recommended practices for handling health-related data and complies with standard data protection policies, promoting trust among users and healthcare partners.

5 **Offline Support and Syncing:**

The Firebase Realtime Database supports offline data persistence, which is particularly beneficial in rural or low-connectivity areas. When the device is offline, data is stored locally and automatically synced with the cloud once the internet connection is re-established. This feature ensures uninterrupted usage and reliability in diverse field conditions.

6 **Scalability and Future-Proofing:**

Firebase can easily scale to handle increasing volumes of users and test data. The backend infrastructure can support thousands of concurrent connections, which is essential for mass screening events. As future updates introduce more complex data types (e.g., voice logs, audio samples), Firebase can be integrated with Cloud Storage or other Google Cloud services to handle extended functionalities without affecting performance.

7 **Integration with Machine Learning Models:**

In the extended version of the project, Firebase acts as a central hub for storing datasets used in training and evaluating machine learning models. The database can store labeled data, predictions, and model feedback loops, enabling continuous model improvement and

tracking of AI performance over time.

8 Analytics and Reporting Support:

With structured data storage, Firebase facilitates integration with tools like Google Data Studio or BigQuery for generating analytical dashboards and performance reports. These insights can be used for academic research, public health policy recommendations, and app feature optimization.

3.4.2 Software Requirements (Platform Choice)

The software development lifecycle of this project involved careful selection of tools, programming languages, and platforms to ensure cross-platform compatibility, ease of development, scalability, and performance. The following subsections detail the key software components used in the development and deployment of the App-Based Audiometer.

1. Operating System:

- **Development Environment:** The application was developed on systems running Windows and macOS, allowing for compatibility testing across multiple platforms. These operating systems provided stable environments for mobile app development, machine learning integration, and database configuration.
- **Deployment Platforms:** The final application is designed to run on both Android and iOS devices, ensuring maximum accessibility and reach among users. The use of Flutter allowed for a single codebase that could be compiled for both platforms, significantly reducing development time and effort while maintaining UI/UX consistency.

2. Programming Languages:

- **Dart:** Dart is the primary programming language used for developing the frontend of the application via the Flutter SDK. Dart offers fast performance, hot reload capabilities, and expressive syntax, making it ideal for developing interactive, real-time mobile applications.
- **Python:** Python was employed for the development and integration of Machine Learning models, including Support Vector Machine (SVM), Random Forest Classifier (RFC), and XGBoost. Python's extensive ML libraries such as scikit-learn and pandas facilitated rapid prototyping, training, and evaluation of predictive models for hearing loss severity classification.

3. Database:

- **Firebase Realtime Database / Firestore:** These cloud-based NoSQL databases were chosen

for secure, scalable, and real-time data storage. The databases store user profiles, hearing test data, audiograms, and ML-generated predictions. Firebase offers built-in user authentication, data synchronization across devices, and robust security features, making it an excellent fit for healthcare-related applications.

4. Development Tools:

- **Android Studio & Visual Studio Code (VS Code):** These were the primary Integrated Development Environments (IDEs) used for building, testing, and debugging the application. Android Studio was particularly useful for emulator testing, APK generation, and performance profiling, while VS Code offered a lightweight and flexible environment for quick code iteration.
- **Flutter SDK:** The Flutter Software Development Kit was central to the entire development process. It provided the widgets, libraries, and APIs needed to build a smooth, high-performance UI for both Android and iOS platforms. Flutter's single codebase architecture helped reduce platform-specific discrepancies and allowed for faster deployment cycles.
- **Python Libraries & Tools:** For machine learning tasks, tools such as Jupyter Notebook, scikit-learn, and NumPy were used for model development and validation. The trained models were later integrated into the mobile application for real-time inference.

3.4.3 Hardware Requirements

The successful development, deployment, and operation of the App-Based Audiometer depend on a variety of hardware components. These components are essential at different stages, including development, testing, user interaction, and data processing. The following hardware resources were employed throughout the project lifecycle:

1. Smartphone (Android/iOS):

Smartphones serve as the primary platform for the application. The app is designed to run efficiently on Android and iOS devices, offering wide accessibility across user demographics. Devices used during testing included mid-range and low-end smartphones to ensure performance consistency across different hardware capabilities. Key requirements include:

- Minimum 2GB RAM for smooth UI rendering.
- Functional Bluetooth (if integrating wireless headsets).

- Microphone access and speaker/headset output support.CPU: Intel Core i3 / AMD Ryzen 3

The app's portability on smartphones allows it to be deployed in rural health camps, school screenings, and workplaces without relying on expensive audiometry hardware.

2. Microphone & Headset:

High-quality earphones or headsets with known frequency response were used to ensure accurate tone delivery during hearing assessments.

CPU: Intel Core i5 / Ryzen 5

- The microphone is used to detect ambient noise levels and perform real-time background noise calibration (ensuring <40 dB noise environment for accurate testing).
- A wired headset is preferred during calibration and test phases to reduce latency and ensure consistent output levels.
- The system uses automated headset calibration routines to match the device output to standardized hearing thresholds.

These components are integral to simulating audiometry hardware in a mobile app environment.

3. Computer/Laptop:

The application development and machine learning model training were conducted on personal computers or laptops with the following minimum specifications:

- Intel i5 or higher processor (or equivalent)
- 8GB RAM
- 256GB SSD (for faster project builds and testing)
- Operating System: Windows 10/11 or macOS (latest stable version)

These systems were used for:

- Writing frontend code (Flutter/Dart)
- Training and testing ML models (Python)
- Version control and project management (Git)
- Emulator-based testing and debugging

The use of high-performance machines helped in reducing build time and improving overall development productivity.

4. Cloud Server (if applicable):

While Firebase serves as the core backend, optional integration with cloud computing resources (e.g., Google Cloud or AWS) allows for:

- Scalable data storage: Storing large volumes of audiograms and user data securely.
- Remote ML inference: In future versions, heavier models could be deployed on the cloud to return predictions without burdening the mobile device.
- Real-time dashboards: Accessible by healthcare professionals and administrators to monitor mass screening campaigns and generate reports.

Using cloud services enhances the reliability, scalability, and performance of the application, especially in large-scale deployment scenarios.

Chapter 4

Proposed

Methodology

The proposed methodology uses a Flutter-based mobile application and an ML model to detect hearing loss. The application uses Firebase as the backend database for storing test results and authentication details. The mobile application features two models: Mass Testing and Individual Testing. When users take the hearing test, their audiograms are stored in the database. After collecting audiograms from 1,000 users, the dataset was processed and analyzed using an RFC, Naive Bayes, SVM and XGBoost to classify hearing loss severity for both the left and right ear separately. Architecture Diagram in Fig 1 illustrates the architecture of the App-Based Audiometer, which consists of two testing models:

- Individual Testing Model :** In the Individual Testing Model, the user logs in using their credentials. Upon signing in, the user can either take a hearing test or view their previous reports. The hearing test begins with a background noise check, where the app measures ambient noise levels and only proceeds if the noise is below 40 dB, ensuring optimal testing conditions. The user then connects their headphones to the device and proceeds to the headset calibration step. To standardize sound levels across different devices, users are prompted to adjust the volume until an 80 dB tone is consistently heard. This ensures uniformity in testing conditions. Once the headset calibration is completed, the app plays tones at various frequencies and intensities for each ear. Users indicate whether they can hear the tone, and based on their responses, an audiogram is generated. The app then classifies the results into three categories: normal hearing, moderate hearing loss, or severe hearing loss, depending on the average intensity levels detected for both ears. Users with an average hearing threshold of ≤ 20 dB are classified as having normal hearing, those with ≤ 50 dB have moderate hearing loss, and those with ≤ 80 dB are categorized as having severe hearing loss.
- Mass Testing Model :** The Mass Testing Model is designed for administrators to monitor large-scale hearing assessments efficiently. In this model, the admin logs in and generates a unique code. This unique code is then distributed to a group of end users who need to take the test. The end users log in via the Individual Testing Model, complete their hearing tests, and upon generating their audiograms, save their results using the unique code provided by the admin. Once the audiograms are saved, the admin can access all submitted results from the group. The admin can filter the data based on hearing loss levels, distinguishing between users with normal hearing and those with hearing impairments. This model is particularly beneficial for organizations conducting hearing assessments in workplaces, schools, or public health initiatives, allowing large-scale data collection and analysis.

- **Machine learning in Audiometry :**

Step 1: Load the Dataset:

The dataset used for training and testing the model consists of hearing test results collected from various users using the proposed ABA. Each data entry includes demographic information such as age and gender, along with hearing test responses recorded at different frequencies and intensity levels. The dataset is stored in a CSV format, which is loaded into a Pandas DataFrame for further processing. By assessing hearing test results, each participant's level of hearing loss in both the left and right ear was determined. The input feature matrix X for each patient i is represented as:

- X be the input feature matrix, where each patient i is represented as:

$$X_i = [L_0, L_1, \dots, L_5, R_0, R_1, \dots, R_5, G, A] \quad (1)$$

- L_j represents the hearing threshold at frequency j for the left ear.
- R_j represents the hearing threshold at frequency j for the right ear.
- G is the gender (binary encoded).
- A is the age.

y_L , and y_R are the target categorical variables representing hearing loss severity for the left and right ears, respectively:

$$y_L, y_R \in \{0(\text{No Loss}), 1(\text{Moderate Loss}), 2(\text{Severe Loss})\} \quad (2)$$

Step 2: Preprocessing the Data:

Before feeding the data into the ML model, several preprocessing steps are applied. The hearing test results, which are stored as string representations of lists, are converted into numerical arrays for analysis. The dataset is then expanded by extracting individual frequency values as separate columns for both left and right ears. The average hearing threshold for each ear is computed as:

$$l_{avg} = \frac{1}{6} \sum_{j=0}^5 L_j \quad r_{avg} = \frac{1}{6} \sum_{j=0}^5 R_j \quad (3)$$

Hearing loss severity is categorized using the function

$$f(x) = \begin{cases} 0, & x \leq 20 \text{ (No Loss)} \\ 1, & 20 < x \leq 50 \text{ (Moderate Loss)} \\ 2, & x > 50 \text{ (Severe Loss)} \end{cases} \quad (4)$$

Standardization is applied:

$$X_{scaled} = \frac{(X - \mu)}{\sigma} \quad (5)$$

where μ is the mean and σ is the standard deviation of each feature.

Categorical variables, such as gender and hearing loss labels, are encoded into numerical values, and all feature variables are standardized to ensure uniformity.

Step 3: Splitting the Dataset into Training and Testing Sets:

To train and evaluate the ML model, the dataset is split into training and testing sets. This ensures that the model is trained on a portion of the data while being evaluated on unseen data, which helps measure its generalization ability. The dataset is divided into an 80:20 ratio, where 80% of the data is used for training and 20% is reserved for testing.

Step 4: Applying the Model:

In ML, a classifier is an algorithm that automatically sorts or categorizes data into one or more "classes." Targets, labels, and categories are all terms used to describe classes.

[A]XGBoost Classifier:

XGBoost optimizes the objective function:

$$L(\theta) = \sum_{i=1}^N \iota(\hat{y}_i, y_i) + \sum_{t=1}^T \Omega(f_t)$$

Where:

- $\iota(\hat{y}_i, y_i)$ is the loss function (log loss for classification).
- $\Omega(f_t)$ is the regularization term for controlling model complexity.
- The final prediction is given by :

$$\hat{y} = \sum_{t=1}^T f_t(X)$$

[B] Support Vector Machine (SVM) Classifier:

SVM finds the optimal hyperplane for classification by solving:

$$\frac{1}{2} ||w||^2 \quad \text{subject to } y_i(w \cdot X_i + b) \geq 1$$

Where:

- W is the weight vector.
- B is the bias term
- y_i is the class label

Kernel function $K(X_i, X_i)$ are used for non-linear classification.

[C] Naïve Bayes Classifier:

Naïve Bayes applies Bayes' theorem:

$$P(X) = \frac{P(X|y)P(y)}{P(X)}$$

Where:

- P(y) is the prior probability of class y.

$$P(X_j|y) = \frac{1}{\sqrt{2\pi\sigma_y^2}} e^{-\frac{(X_j - \mu_y)^2}{2\sigma_y^2}}$$

The final prediction is:

$$\hat{y} = \arg \arg P(y|X)$$

(D) RFC :

classifier is an ensemble of N decision trees:

$$F(X) = \frac{1}{N} \sum_{i=1}^N T_i(X) \quad (6)$$

where each tree $T_i(X)$ predicts a class label based on feature thresholds.

Step 5: Model Evaluation:

After training, the model is evaluated on the test dataset to measure its accuracy and performance. The trained model predicts hearing loss classifications for the test set, and these predictions are compared with actual labels to assess accuracy.

The model optimizes the Gini Impurity at each decision node:

$$G = 1 - \sum_c P(c)^2 \quad (7)$$

where P(c) is the probability of class c in a node.

The final prediction is obtained using majority voting across trees. Evaluation metrics such as accuracy score and classification reports (precision, recall, and F1-score) provide insights into how well the model distinguishes between different levels of hearing loss.

4.1 Proposed System Architecture/Block Diagram

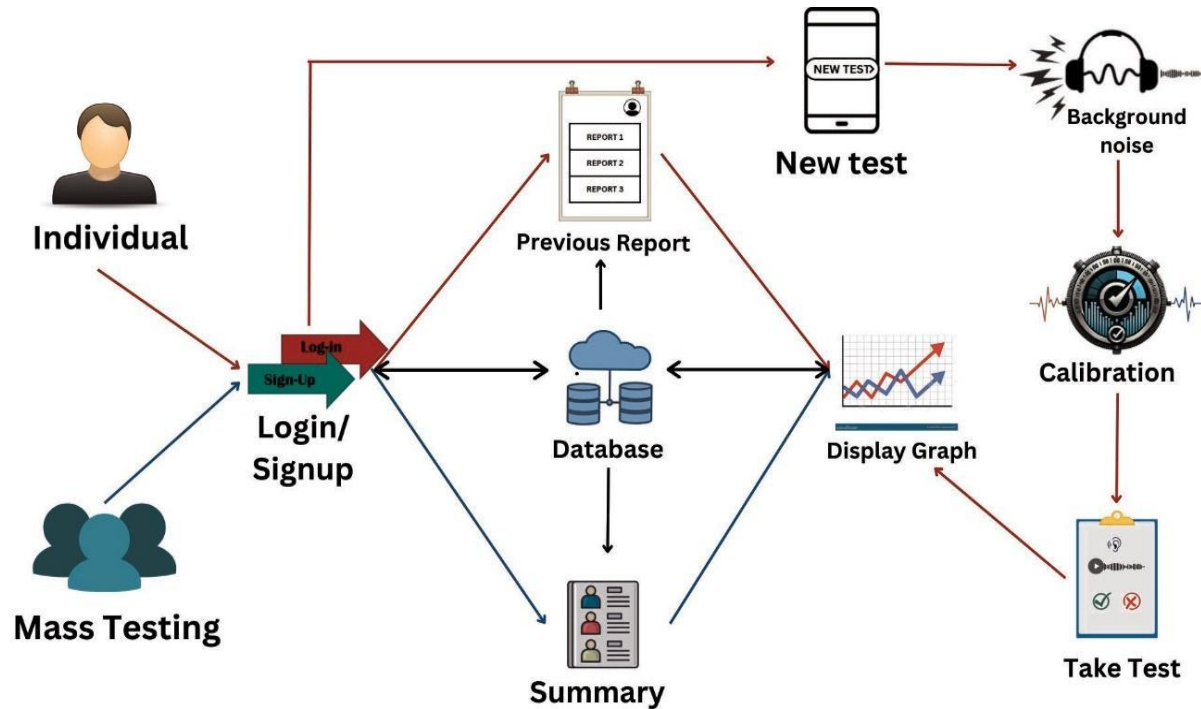


Fig 4. 1 Architecture of App-Based Audiometer

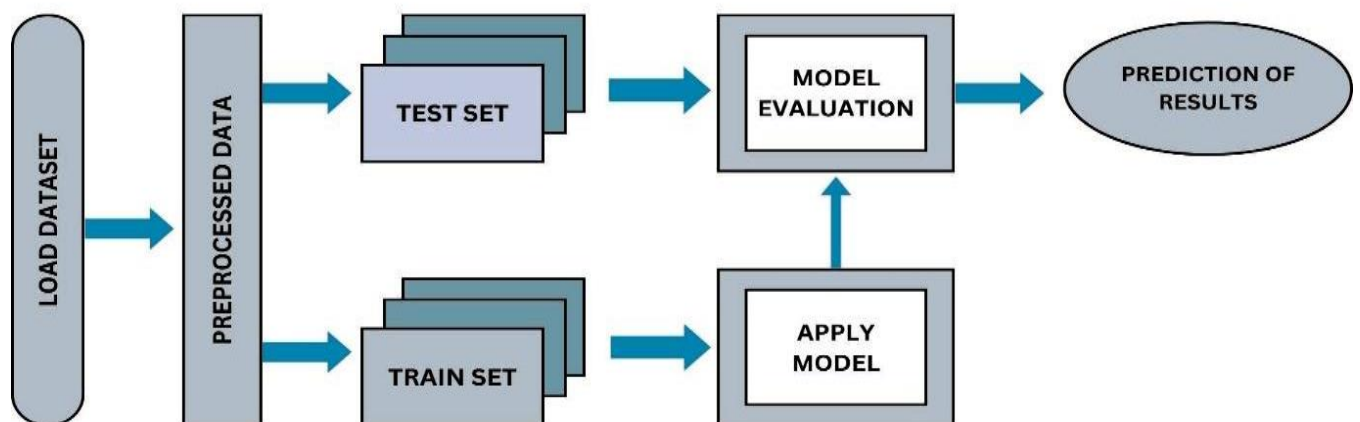


Fig 4.2 Flowchart diagram of ML Model

4.2 Dataset/ Database design:

The overall dataset collected consisted of 1000 users. Initially, data from 25 people was validated by an ENT clinic to ensure accuracy between traditional audiometry tests and proposed ABA tests. After validating the proposed app, the remaining data was collected from various locations, including gardens(20%), colleges(10%), society(20%), hospitals(8%), public campaigns(32%) and others(10%). The dataset included parameters such as age, gender, location, and detailed test results. The hearing test recorded the minimum intensity at which a user could perceive tones at different frequencies. The frequency levels tested were 250 Hz, 500 Hz, 1000 Hz, 2000 Hz, 4000 Hz, and 8000 Hz, while the intensity levels ranged from 10 dB to 80 dB in increments of 10 dB. The dataset for the app-based digital audiometer project was collected in two phases to ensure accuracy and reliability. Initially, data from 25 users were validated at an ENT clinic by comparing results from traditional audiometry tests with those obtained through the proposed ABA (App-Based Audiometry) test. This validation phase confirmed the app's accuracy before expanding data collection.

In the second phase, data from 975 additional users were gathered from diverse locations to ensure a well-distributed sample. Public campaigns contributed the highest share, accounting for 32% of the dataset, followed by gardens and parks at 20%, residential societies at 20%, colleges at 10%, hospitals at 8%, and other locations such as offices and local events at 10%. Each participant underwent a hearing test in a relatively quiet environment, where the app recorded the minimum intensity (in dB) at which they could detect tones at six key frequencies: 250 Hz, 500 Hz, 1000 Hz, 2000 Hz, 4000 Hz, and 8000 Hz. The data included individual test results for both ears, along with demographic information such as age, gender, and location. The hearing loss classification was determined by calculating the average hearing threshold for each ear and categorizing it as No Loss, Moderate Loss, or Severe Loss.

This dataset provides a diverse and realistic representation of hearing abilities across different demographics, making it valuable for further analysis and improvement of hearing loss detection methods.

4.3 Overview of Project Modules:

The **App-Based Digital Audiometer** is developed with a modular architecture to ensure flexibility, maintainability, and scalability. Each module is designed to serve a specific purpose in the hearing test and analysis pipeline, while collectively delivering a seamless, accurate, and user-friendly experience. Below is a detailed explanation of each core module within the application:

1 User Registration and Profile Management

This module acts as the entry point for users interacting with the app. It allows individuals to create personal profiles and access their hearing test history.

- **Sign-Up and Profile Creation:** Users are required to input their name, age, gender, location, and contact details. This information is crucial for demographic analysis and personalized record keeping.
- **Login and Authentication:** Integrated with Firebase Authentication, this module ensures secure and quick user login, while maintaining privacy.
- **Profile Management:** Users can update their personal information, review their previous hearing test records, and monitor progress over time. The system also supports the creation of profiles for dependent users, such as elderly family members or children.
- **Accessibility Features:** Designed with a user-centric UI that supports ease of use for individuals of all age groups, especially the elderly.

2 Hearing Test Module

The core functionality of the application resides in this module. It replicates traditional pure-tone audiometry through a mobile-based interface.

- **Test Initialization:** Upon starting the test, the user is guided through simple instructions. The UI features clear icons and voice prompts to help users understand the steps involved.
- **Frequency Tones Playback:** The module plays audio tones at six standard frequencies (250 Hz, 500 Hz, 1000 Hz, 2000 Hz, 4000 Hz, and 8000 Hz).
- **Volume Level Detection:** For each frequency, the intensity is gradually increased from 10 dB to 80 dB. The app records the minimum intensity at which the user can detect the tone for both ears.
- **Result Logging:** The perceived intensity values are captured and passed to the processing module for further analysis.

- **Real-Time Feedback:** Users receive on-screen prompts after each tone, helping maintain engagement and accuracy.

3 Data Collection and Processing:

This module is responsible for capturing, storing, and processing raw test data.

- **Data Storage:** Collected test results (intensity vs. frequency per ear), along with user demographic and location data, are securely stored in Firebase Firestore.
- **Threshold Computation:** The app computes average hearing thresholds for each ear and determines the overall auditory capability.
- **Hearing Loss Classification:** Based on predefined thresholds, the system categorizes hearing loss into three levels:
 - a. No Loss
 - b. Moderate Loss
 - c. Severe Loss
- **Anomaly Checks:** If irregular test patterns are detected (e.g., unresponsiveness or illogical responses), the system flags the test for retake.

4 Validation and Comparison Module

To ensure medical reliability, this module is integrated for validation of app-based results against conventional audiometry.

- **Clinic Validation:** Data from 25 individuals was first cross-validated with results from an ENT clinic to confirm the accuracy of the app.
- **Statistical Analysis:** Python scripts were used to conduct paired t-tests and correlation analysis to compare traditional and app-based results.
- **Ongoing Calibration:** The app is periodically calibrated based on ongoing comparison with standard audiometric data.

5 Results Analysis and Visualization

This module is designed to convert raw data into meaningful insights for both users and healthcare providers.

- **Graphical Representation:** Hearing test results are shown through bar graphs or line charts for each frequency, helping users visually understand their hearing profile.
- **Trend Monitoring:** Users can track changes in hearing ability over time, especially useful for monitoring chronic hearing degradation.
- **Medical Recommendations:** The system flags cases of Moderate or Severe Loss and prompts users to consult an audiologist.
- **Downloadable Reports:** Users can export their hearing reports in PDF format for medical use.

6 Database and Cloud Storage Module

This module ensures secure, scalable, and structured data storage.

- **Cloud Integration:** Utilizes Firebase Firestore to store user profiles, hearing test data, and classification results.
- **Data Privacy and Security:** Data is encrypted and stored in compliance with health-tech security standards (e.g., HIPAA-style encryption logic where applicable).
- **Scalability:** The database is designed to handle increasing users and large volumes of test results with minimal latency.

7 Admin and Monitoring Module

This administrative interface is built for researchers, ENT professionals, and project supervisors to monitor and manage the entire dataset.

- **Access Control:** Only authorized personnel can access this module through secure login credentials.
- **Dataset Monitoring:** Provides dashboards to view test trends across age groups, locations, and time periods.

- **Insight Generation:** Offers analytical tools to generate reports on population-level hearing health, identify at-risk demographics, and evaluate the performance of screening campaigns.
- **Feedback Loop:** Doctors and researchers can provide feedback on test outcomes, contributing to iterative improvements in the app's algorithm and UI.

The **App-Based Digital Audiometer** is a comprehensive, modularly designed hearing assessment tool that brings clinical-level audiometry testing into the hands of users through a smartphone application. Each module—ranging from user registration to real-time analysis and cloud-based storage—contributes to building a holistic system that is accurate, scalable, and easy to use. The modular architecture ensures that each component functions independently while contributing to the overall synergy of the app, making it easier to test, maintain, and enhance over time. The **User Registration and Profile Management module** ensures personalized interaction by allowing users to create and manage profiles, track historical test data, and even manage dependent accounts for family members. This feature is especially useful for screening children or elderly individuals. The **Hearing Test module** brings audiometric precision to mobile platforms by using standardized pure-tone frequencies and intensity levels. It provides an interactive testing experience, guided by visual and auditory cues, that replicates traditional audiometry in a much more accessible format. All collected data is seamlessly handled by the **Data Collection and Processing module**, which stores raw frequency-intensity information along with demographic details. It processes this information to compute hearing thresholds and categorizes hearing health into actionable levels—No Loss, Moderate Loss, or Severe Loss—making it easy to interpret even for non-medical users. To ensure medical credibility, the **Validation and Comparison module** was implemented early in the development cycle. This module compared app results against ENT clinic data, applying statistical validation through Python-based scripts to ensure the reliability of the test outcomes. The **Results Analysis and Visualization module** turns complex audiometric data into intuitive charts and reports, enabling both users and healthcare professionals to understand and act on the results. Features like downloadable PDFs, trend tracking, and health recommendations further increase the app's utility.

Data integrity and privacy are managed through the **Database and Cloud Storage module**, which uses Firebase for secure, scalable storage while ensuring that health data is encrypted and accessible only by authorized users.

Lastly, the **Admin and Monitoring module** provides powerful tools for researchers, doctors, and

project stakeholders. It includes dashboards, trend analysis tools, and dataset access for public health campaigns and research purposes.

Collectively, these modules form a powerful ecosystem for digital hearing assessment. Future enhancements are already being envisioned, including AI-driven hearing loss prediction, real-time syncing with telemedicine services, and multilingual support for outreach in rural and underserved communities. This structured and forward-compatible design positions the App-Based Digital Audiometer as a transformative tool in preventive auditory healthcare.

4.4 Tools and Technologies Used:

The development of the App-Based Digital Audiometer leveraged modern technologies to ensure accuracy, scalability, and efficiency. The key tools and technologies used in the project are:

1. **Flutter :**

- a) Used for developing a cross-platform mobile application with a smooth and responsive UI.
- b) Provides a consistent user experience across both Android and iOS devices.
- c) Enables real-time interaction and audio playback for hearing tests.

2. **Firestore :**

- a) Serves as the backend for authentication, database management, and cloud storage.
- b) Firestore stores user profiles and hearing test results securely.
- c) Firebase Authentication handles user login and access control.

3. **Python :**

- a) Used for building model , analysis and validation of audiometry test results.
- b) Implements statistical comparison between app-based hearing tests and traditional audiometry tests.
- c) Enables advanced data processing for identifying hearing loss trends.

4. **GitHub :**

- a) Used for version control and collaborative development.
- b) Facilitates efficient code management and tracking of feature updates.
- c) Ensures a structured development workflow with branches, commits, and pull requests.

4.5 Entity Relationship Diagram:

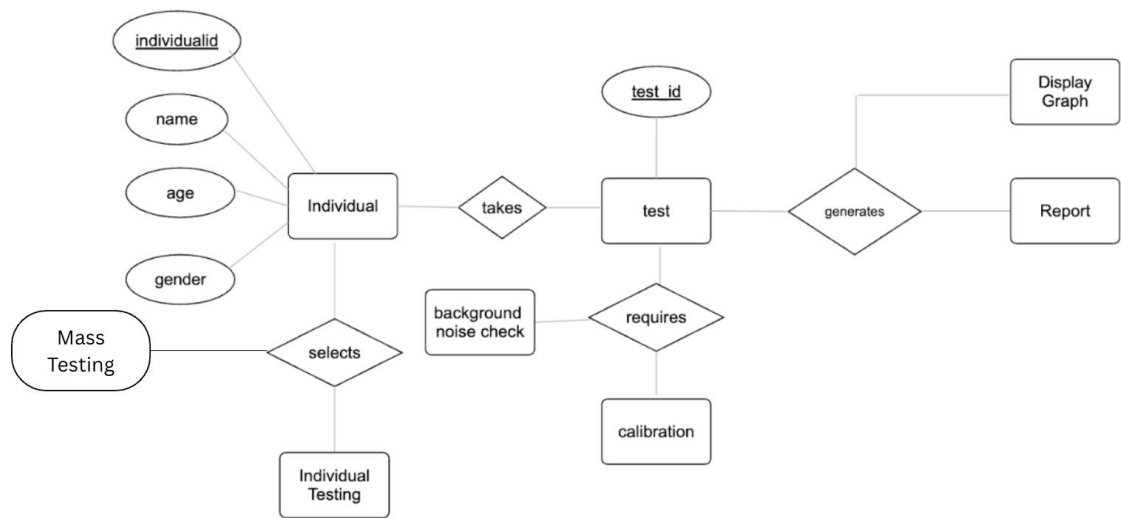


Fig 4.3 Entity Relationship Diagram

4.6 Use Case Diagram:

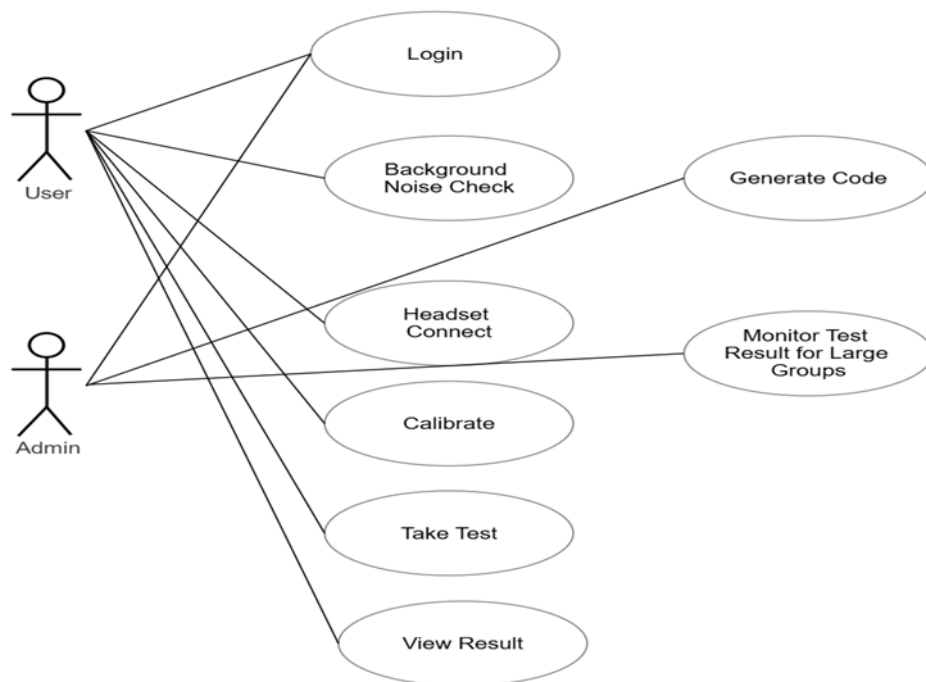


Fig 4.4 Use Case Diagram

4.7 Timeline UML Diagram:

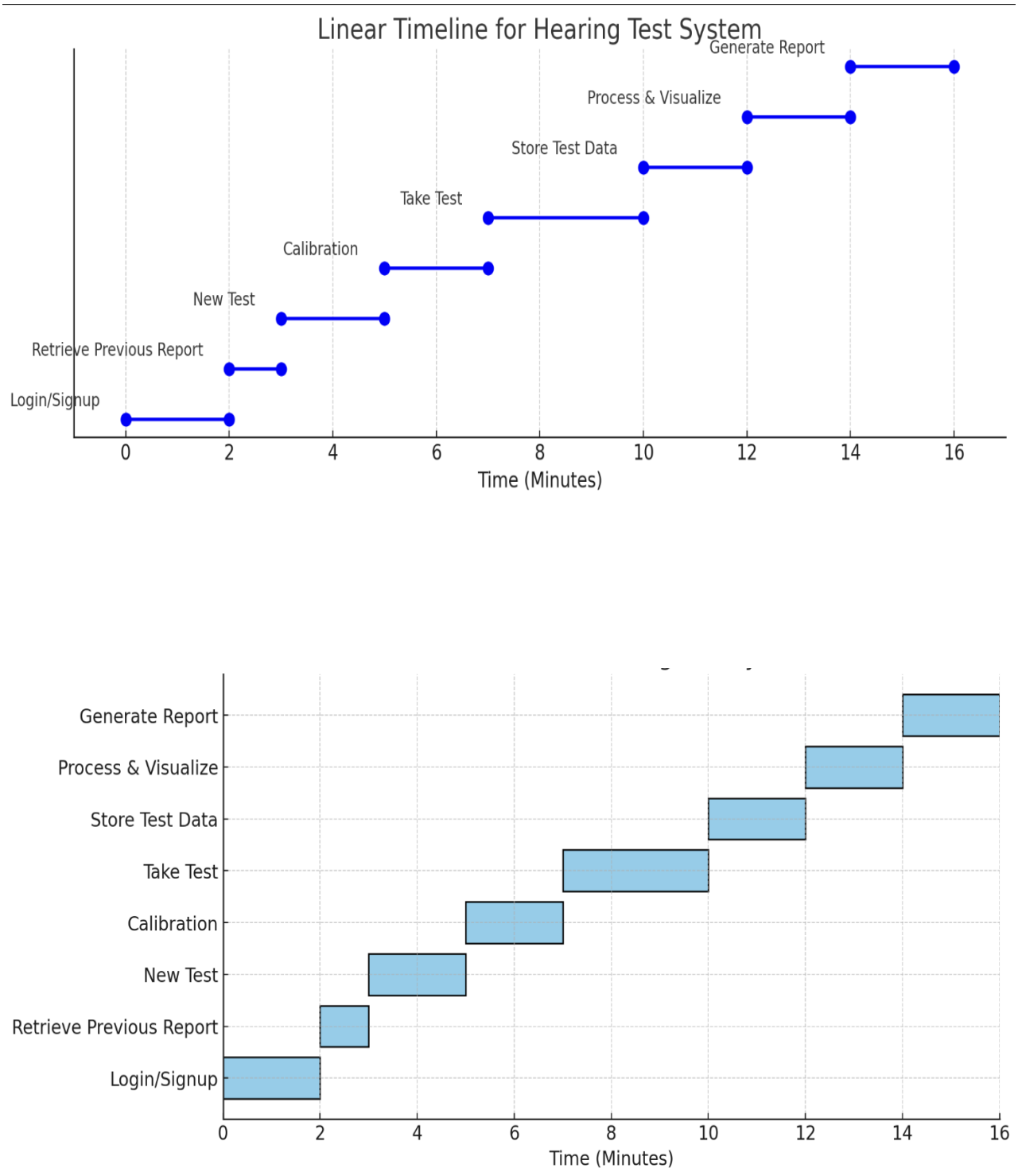


Fig 4.5 Timeline UML Diagram

4.8 SDLC Model Applied: Agile Methodology:

The Agile Software Development Life Cycle (SDLC) was adopted for the development of the App-Based Digital Audiometer to ensure iterative progress, continuous user feedback integration, and a flexible development environment. Agile proved to be the most suitable model for this project due to its ability to handle changing requirements and incorporate regular improvements through short development cycles, known as sprints.

Why Agile?

- The nature of the project required frequent testing and validation, especially in medical and field conditions.
- Feedback from ENT specialists, users, and field testers had to be incorporated dynamically.
- Features evolved based on data insights and real-world usage, making Agile the perfect fit.

Key Agile Phases and Their Execution in the Project

1) Requirement Gathering & Planning

- a) Conducted brainstorming sessions with mentors, domain experts, and potential users (ENT doctors, school screening personnel).
- b) Identified core objectives: accurate hearing testing, offline and mobile accessibility, intuitive UI, and easy data management.
- c) Broke down the entire project into user stories and tasks, prioritizing them based on criticality and feasibility.
- d) Defined the technology stack: Flutter for the frontend, Firebase for backend services, Python for analytics, and GitHub for collaboration.

2) Design & Prototyping

- a) Designed low-fidelity wireframes for user onboarding, hearing test interface, and result dashboards.
- b) Created a clickable prototype using Flutter to simulate the flow and functionality.
- c) Finalized the UI/UX design focusing on accessibility, especially for elderly users.
- d) Developed a database structure in Firebase to support real-time storage and retrieval of test data.

3) Incremental Development & Sprint Cycles

- a) The development was broken into 5 sprint cycles, each lasting around 1–2 weeks:
- b) Sprint 1: Basic user registration, authentication using Firebase, and UI layout for home screen.
- c) Sprint 2: Core hearing test module – frequency generation, intensity control, and user response capture.
- d) Sprint 3: Data storage and classification module – calculating average thresholds and categorizing hearing loss.
- e) Sprint 4: Visualization – graphical representation of results and history tracking.
- f) Sprint 5: Testing, optimization, bug fixes, and preparation for field testing.
- g) Regular stand-up meetings were held to track progress, resolve blockers, and reprioritize tasks.

4) Testing & Validation

- a) Functional testing was done after each sprint to ensure the module met expectations.
- b) Validated the app's accuracy by comparing results from 25 initial users with traditional audiometry results at an ENT clinic.
- c) Performed field testing across diverse environments like societies, gardens, and hospitals to test the app in real-world conditions.
- d) User feedback was actively collected and documented during these tests to guide future enhancements.

5) Deployment & Maintenance

- a) After successful validation, the app was deployed to mobile devices for large-scale data collection.
- b) Firebase handled backend services, ensuring seamless real-time data updates and centralized storage.
- c) GitHub was used for continuous integration and version tracking, allowing multiple contributors to collaborate effectively.
- d) A lightweight admin module was also created to view and analyze collected test data for research and monitoring.

6) Continuous Improvement & Future Scope

- a) Agile promotes continuous delivery, allowing us to keep refining the application with each cycle.
- b) Future iterations include:
- c) Integration of AI models for automatic hearing loss prediction.
- d) Voice command-based interaction for visually impaired users.
- e) Expanded multilingual support for rural and regional users.
- f) Detailed analytics dashboards for healthcare professionals.

Chapter 5

Project Plan

5.1 Risk Management

Risk management is a crucial part of the App-Based Audiometer project to ensure smooth execution and delivery. It involves identifying potential risks, analyzing their impact, and developing mitigation strategies. Effective risk management minimizes uncertainties and ensures the project stays on track.

5.1.1 Risk Identification

In this section, we identify potential risks that may arise during the **development, testing, and deployment** of the project. These risks are categorized into four main types:

1. Technical Risks

These risks are related to technological challenges and the performance of the system.

- **Inaccurate ML predictions:** The machine learning model used for hearing loss detection might produce inaccurate results due to insufficient or imbalanced training data, overfitting or underfitting, and poor feature selection. This can reduce the reliability and accuracy of the app, making it less effective in detecting hearing loss. Inaccurate predictions can lead to false positives or false negatives, which could affect the trustworthiness of the app and compromise the users' health assessments.
- **Real-time data synchronization issues:** Since the project uses Firebase Realtime Database for storing and synchronizing data, potential issues may arise. Network latency or unstable internet connections could cause delays in data retrieval and updates, making the app's results inconsistent. Additionally, unreliable data synchronization may lead to inaccurate test outcomes, as the app relies on real-time updates for consistent performance.
- **Compatibility issues with devices:** The app might face compatibility issues with specific smartphone models or operating systems. Differences in hardware configurations, such as

microphone sensitivity or headphone quality, may lead to inconsistent hearing test results. This variation could reduce the app's accuracy and reliability across different devices, limiting its accessibility and effectiveness.

2. Operational Risks

These risks arise from project execution and day-to-day operations.

- **Hardware unavailability:** Testing the app requires headphones and smartphones with specific specifications to maintain the accuracy of hearing assessments. If the required hardware is unavailable during testing, it may delay the validation phase. Without proper hardware, the team cannot conduct comprehensive testing, increasing the risk of undetected bugs or inaccuracies.
- **Delay in testing and debugging:** Issues in testing and debugging could slow down the project's progress. If bugs are detected late in the development cycle, it may lead to a bottleneck that requires significant rework. This can affect the overall timeline, pushing back milestones and delivery dates. Incomplete testing could also result in undetected errors, impacting the app's quality and performance.

3. Project Management Risks

These are risks associated with project planning, resource allocation, and scope management.

- **Scope creep:** The project might face scope creep if additional features are requested or new requirements emerge during development. This could lead to an increase in development time and resource utilization, potentially causing the team to miss deadlines. Scope creep may also introduce unexpected technical challenges, making the project more complex and difficult to manage.
- **Failure to meet deadlines:** Delays in any phase, such as development, testing, or deployment, could cause the project to miss its predefined milestones. This could result in rushed implementation, potentially compromising the quality of the final product. Failure to meet deadlines may also affect the team's reputation and stakeholder satisfaction.

4. Security Risks

These risks pertain to data privacy, security vulnerabilities, and protection of sensitive user data.

- **Data privacy concerns:** The app collects sensitive hearing test data, which may raise privacy concerns. Without proper encryption or security measures, the app is vulnerable to unauthorized access or data breaches. Such incidents could compromise user confidentiality and erode trust in the product. Maintaining strict data privacy standards is essential to avoid legal and ethical issues.
- **Firebase database security issues:** Inadequate authentication and authorization mechanisms in Firebase could expose the database to potential exploitation. Unauthorized access to the database may lead to data tampering or leakage of users' hearing test results. This could compromise the integrity and privacy of the app, making it susceptible to security vulnerabilities.

5.1.2 Risk Analysis

Risk analysis involves evaluating the likelihood and impact of each identified risk on the App-Based Audiometer project. This section assesses the severity and potential consequences of the risks to prioritize them and determine appropriate mitigation strategies. The risks are categorized by probability (Low, Medium, High) and impact (Minor, Moderate, Severe) using a risk matrix.

5.1.3 Overview of Risk Mitigation, Monitoring, Management

Table 5.1 Risk Mitigation, Monitoring and Management

Risk	Probability	Impact	Severity Level
Inaccurate ML predictions	Medium	Severe	High
Real-time data sync issues	High	Moderate	High
Compatibility issues with devices	Medium	Moderate	Medium
Hardware unavailability	Low	Moderate	Low
Delay in testing and debugging	Medium	High	High
Scope creep	Medium	Moderate	Medium
Failure to meet deadlines	Medium	Severe	High
Data privacy concerns	High	Severe	High
Firebase security issues	Medium	Severe	High

5.1.3.1 Risk Mitigation

Mitigation refers to the strategies and actions taken to reduce the likelihood or impact of potential risks. It focuses on proactively preventing risks from materializing or minimizing their consequences.

- **Technical Risk Mitigation:**

To address inaccurate ML predictions, the model is trained with larger, diverse, and balanced datasets to enhance accuracy. Continuous model fine-tuning is performed to prevent overfitting and underfitting. Additionally, regular testing and validation of the model with real-world data are conducted to improve its reliability. To mitigate real-time data synchronization issues, Firebase's synchronization mechanisms are optimized, and offline caching is introduced. This ensures that data retrieval and updates are consistent, even in cases of network instability. Compatibility risks are minimized by thoroughly testing the app on a variety of Android and iOS devices with different hardware configurations, ensuring stable performance across platforms.

- **Operational Risk Mitigation:**

Hardware unavailability is mitigated by ensuring that all necessary devices, such as smartphones and headphones, are procured before the testing phase. Backup devices are kept on hand to avoid delays. To prevent delays in testing and debugging, buffer time is incorporated into the project schedule. Automated testing frameworks and continuous integration practices are applied to detect and resolve issues early, reducing the risk of major setbacks.

- **Project Management Risk Mitigation:**

Scope creep is controlled by clearly defining and documenting project requirements upfront. Any new features or changes undergo a formal review and approval process to prevent uncontrolled expansion. To minimize the risk of missing deadlines, realistic milestones are established, and project management tools are used to track progress. Regular sprint reviews are conducted to identify and address potential delays early.

- **Security Risk Mitigation:**

To address data privacy concerns, sensitive hearing test data is encrypted to prevent unauthorized access. Robust authentication and access control mechanisms are implemented in Firebase to safeguard user data. Security audits and penetration testing are performed periodically to identify and address vulnerabilities. This ensures the app maintains strong protection against data breaches.

5.1.3.2. Risk Monitoring

Monitoring refers to the continuous observation and assessment of identified risks throughout the project lifecycle. This ensures that risks are tracked, evaluated, and addressed promptly.

- **Regular Risk Assessments:**

Regular reviews are conducted to monitor the status of existing risks and identify any emerging threats. The project team holds frequent meetings to assess whether current mitigation strategies are effective or need adjustments.

- **Risk Tracking and Logging:**

A risk log is maintained to document all identified risks, their severity, and the corresponding mitigation measures. This log helps the team prioritize high-impact risks and track their evolution over time. Automated monitoring tools are integrated into Firebase to detect synchronization issues and security vulnerabilities in real time. Performance metrics, such as synchronization success rates and app stability, are tracked to identify deviations.

- **Continuous Feedback:**

Beta testers are regularly consulted to gather feedback on app performance and identify potential usability issues. Their insights help the team detect hidden risks and refine the app accordingly. Error-handling logs are automatically generated to capture and analyze technical faults, enabling faster troubleshooting and reducing downtime.

5.1.3.3. Risk Management

Risk management is the overarching process of identifying, assessing, mitigating, and monitoring risks throughout the project. It ensures that the project remains resilient and adaptable in the face of uncertainties.

- **Risk Register and Documentation:**

A detailed risk register is maintained throughout the project lifecycle. It contains information on identified risks, their severity, mitigation strategies, and the responsible team members. This documentation ensures accountability and allows the team to track the effectiveness of risk-handling measures.

- **Version Control and Change Management:**

The project uses version control systems, such as GitHub, to track code changes. This helps identify the source of technical issues early and facilitates rollback if necessary. Proper change management practices ensure that modifications are thoroughly tested before deployment, minimizing the introduction of new risks.

- **Effective Communication:**

The team uses collaboration platforms (e.g., Slack or Microsoft Teams) to share real-time updates on risks and their status. Regular team meetings and sprint reviews facilitate transparency, ensuring that all members are aligned on risk-related activities. Stakeholders are kept informed about major risks and their mitigation plans to ensure proper decision-making and resource allocation.

- **Contingency Plans:**

For critical risks, contingency plans are established to minimize potential disruption. For example, in the event of data synchronization failures, offline data caching serves as a backup mechanism to prevent data loss. Similarly, backup devices and additional testing environments are kept ready to avoid delays caused by hardware unavailability or technical faults.

5.2 Project Schedule

The project was managed using the **Agile methodology**, specifically the **Scrum framework**, which emphasizes iterative development, adaptability, and continuous feedback. The development cycle was divided into **nine sprints**, each spanning approximately one to two weeks. The team conducted regular sprint planning, daily stand-ups, reviews, and retrospectives to ensure that tasks

aligned with the goals of the sprint and the overall vision of the project.

This approach allowed the team to remain flexible, incorporate feedback from real-world testing, and adapt quickly to challenges that arose during the development cycle.

5.2.1 Project Task Set

- **Sprint 1: Initiation and Planning**

The first sprint set the foundation for the entire project. It began with a thorough analysis of the problem statement—mass screening and early detection of hearing loss through an affordable, scalable digital solution. The team discussed the project scope and aligned it with real-world constraints such as cost, device compatibility, and target user group. A feasibility analysis helped evaluate technical risks, required technologies (like Flutter and Firebase), and resource availability. Each team member was then assigned a specific role based on their strengths—for example, UI design, backend development, testing, and documentation.

- **Sprint 2: Design and Prototyping**

This sprint focused on user experience and visual design. Using Figma, detailed wireframes were created for the app's key functionalities including the home screen, hearing test screens, result pages, and authentication flow. A user journey was mapped to visualize how different types of users (e.g., therapists, patients) would navigate through the app. The team also created an interactive prototype in UiZard, which allowed stakeholders to preview the app's basic flow and provide early feedback on usability and design choices.

- **Sprint 3: Flutter App Initialization and UI Setup**

In Sprint 3, the Flutter project was initialized, and core routing/navigation among screens was implemented. Basic UI screens were built to match the wireframes from Sprint 2. The team utilized the VelocityX framework, which streamlined UI development by providing prebuilt responsive components and a visually appealing layout system. This sprint established the app's front-end backbone and helped developers visualize screen transitions and initial layouts.

- **Sprint 4: Firebase and Backend Integration**

This sprint marked the beginning of backend integration. The team configured Firebase Realtime Database to store and retrieve user data in real time. Google Sign-In authentication was implemented for secure access, reducing login friction for users. Furthermore, a robust data schema was designed to manage user profiles, hearing test results, and app configurations. This sprint established the basic cloud connectivity and database model that future modules would build upon.

- **Sprint 5: Audiometer Logic and Data Capture**

The core functionality of the app—audiometric testing—was developed in this sprint. The app was programmed to generate sound tones of specific frequencies (ranging from 250 Hz to 8000 Hz) and intensities for both ears. A calibration UI was implemented to allow users to indicate when they could hear a tone, and based on their inputs, thresholds were calculated. The data from each test session was then securely saved into Firebase for further analysis. This sprint was crucial as it translated clinical testing protocols into a usable app feature.

- **Sprint 6: Machine Learning Model Integration**

This sprint focused on integrating a Machine Learning-based hearing classification system into the app. The data collected from user audiometry sessions—comprising hearing thresholds across multiple frequencies—was utilized to train various ML classifiers such as Support Vector Machine (SVM), Random Forest (RF), and XGBoost. These models were evaluated to determine which achieved the highest classification accuracy for identifying the severity of hearing loss: no loss, moderate, or severe. Additionally, we compared the sensitivity and specificity of our ML predictions to assess how accurately the system could differentiate between hearing and non-hearing individuals. A comparative analysis was also carried out between results from traditional audiometers and our app-based approach to validate the reliability of the system. While the models were tested locally, the architecture was designed to support future cloud-based deployment for scalable, accessible, and efficient hearing diagnosis.

- **Sprint 7: Testing and Performance Optimization**

The focus in this sprint shifted to quality assurance. Unit testing was conducted using Mockito, FirebaseAuthMock, and FirebaseDatabaseMock to ensure that individual components worked correctly without relying on live Firebase data. This allowed safer and

faster testing. Additionally, integration testing verified that modules like authentication, data storage, and ML predictions worked seamlessly. Performance profiling was also done to optimize memory usage, load times, and responsiveness, which is critical for real-time testing applications.

- **Sprint 8: Pilot Testing and Feedback Loop**

This sprint involved real-world pilot testing of the app with a small group of users, including speech-language therapists and test subjects. Feedback was collected regarding the ease of use, audio clarity, accuracy of results, and overall app experience. Based on this feedback, UI tweaks were made for better accessibility, and minor bugs were fixed. This iterative improvement based on real user input helped enhance the overall reliability and usability of the app.

- **Sprint 9: Final Touches and Documentation**

The final sprint focused on wrapping up the project. Although deployment was not conducted in this project phase, the team ensured the app was well-tested and submission-ready. All technical documentation was compiled, including system architecture, codebase explanations, data schemas, and user manuals. These artifacts were designed to support future development, deployment, or submission for funding, hackathons, or institutional reviews.

5.2.2 Timeline Chart

The project followed an Agile methodology, broken into 9 structured sprints spread across a 20-week timeline. Each sprint focused on specific modules and user stories, allowing iterative development, continuous feedback, and refinement. The following Gantt chart outlines the sprint schedule and major tasks over the project duration.

Table 5.2 Gantt Chart Representation:

Week	Sprint	Task Highlights
1–2	Sprint 1	Defined project scope, problem statement, feasibility analysis, resource planning, team roles
3–4	Sprint 2	UI/UX wireframe design in Figma, user journey mapping, initial prototype using Ui-Zard
5–6	Sprint 3	Flutter project setup, navigation and UI screen development, VelocityX UI integration
7–8	Sprint 4	Firebase Realtime Database setup, Google Sign-In authentication, data schema and model design
9–10	Sprint 5	Audiometer logic implementation, tone generation, threshold calibration UI, Firebase integration
11–13	Sprint 6	ML model development for hearing classification, test data evaluation, sensitivity/specificity
14–15	Sprint 7	Unit testing using Mockito, firebase_auth_mock, integration testing, performance optimization
16–17	Sprint 8	Pilot testing with real users, feedback collection, bug fixes, UI refinements
18–20	Sprint 9	Final testing, documentation, user manual, and submission preparation

5.3 Sustainability of Project:

The sustainability of the App-Based Audiometer for Hearing Loss Detection and Prediction Using Machine Learning focuses on ensuring its long-term viability, scalability, and adaptability. It addresses economic, environmental, social, and technical sustainability, ensuring the project continues to deliver value while being cost-effective, eco-friendly, and beneficial to society. Moreover, the project aligns with the United Nations Sustainable Development Goal (SDG) 3: Good Health and Well-being, which aims to ensure healthy lives and promote well-being for all ages by enhancing access to healthcare through early detection and preventive measures.

1. Economic Sustainability:

The project is designed to be cost-effective by using readily available resources such as smartphones, headphones, and Firebase for cloud storage. This makes the solution affordable for both individual users and healthcare providers.

- **Low-Cost Deployment:**

App Based Audiometer for Hearing Loss Detection The app can be deployed with minimal infrastructure requirements, making it suitable for widespread adoption in both urban and rural areas.

- **Scalable Business Model:**

The project has the potential for commercial scalability by offering the app as a subscription-based service or partnering with healthcare organizations.

- **Future Revenue Opportunities:**

- Licensing the app to hospitals, clinics, and audiology centers.
- Integration with telehealth platforms for remote consultations.
- Expansion into global markets by adapting the app for multiple languages and regions.

2. Environmental Sustainability:

The app promotes eco-friendly practices by leveraging digital technology to reduce the need for physical audiometry equipment.

- **Reduced E-Waste Generation:**

Traditional audiometry devices are costly and contribute to electronic waste over time. By using smartphones and headphones, the app reduces the need for specialized hardware, thereby lowering e-waste production.

- **Paperless Operations:**

The app uses digital reports and cloud storage, eliminating the need for physical documentation, contributing to paper conservation.

- **Energy Efficiency:**

The Firebase Realtime Database used for cloud storage ensures optimized data handling, reducing the need for large-scale servers, which consume more energy.

3. Social Sustainability:

The project contributes to public health awareness and promotes early detection of hearing loss, leading to better health outcomes. This directly supports SDG 3: Good Health and Well-being by improving access to preventive healthcare.

- **Accessible Hearing Tests:**

The app provides affordable and accessible hearing assessments to individuals who may not have access to audiology clinics. This is particularly beneficial in remote and underserved areas, addressing healthcare inequality.

- **User-Friendly Interface:**

The app is designed to be easy to use for all age groups, including elderly individuals who may require regular hearing checks.

- **Promoting Preventive Healthcare:**

By enabling early detection of hearing issues, the app promotes preventive healthcare practices, helping users seek timely medical intervention.

- **Contribution to SDG 3:**

The app helps in reducing the prevalence of untreated hearing loss by enabling early intervention, thereby supporting the goal of ensuring healthy lives and promoting well-being.

4. Technical Sustainability:

The app is built using scalable and maintainable technologies, ensuring long-term technical viability.

- **Scalable Cloud Infrastructure:**

The Firebase Realtime Database provides real-time synchronization and can handle increasing amounts of user data as the project scales.

- **Modular Architecture:**

The app is designed with a modular architecture, making it easy to add new features or update existing ones without disrupting the entire system.

- **Continuous Updates and Maintenance:**

The app can be regularly updated to include new machine learning models, improved accuracy, and enhanced user experience.

- **Cross-Platform Compatibility:**

The app's flexibility allows it to be adapted for iOS, Android, and web platforms, ensuring broad accessibility.

5. Long-Term Impact and Future Enhancements:

The sustainability of the project also depends on its continuous improvement and expansion.

- **Integration with Wearable Devices:**

In the future, the app can be integrated with wearable devices (e.g., smart hearing aids, fitness trackers) for continuous hearing monitoring.

- **AI-Powered Diagnostics:**

Enhancing the machine learning model with AI-powered diagnostic capabilities could enable the app to identify specific hearing conditions with greater accuracy.

- **Collaboration with Healthcare Providers:**

Forming partnerships with hospitals and audiologists can enhance credibility and facilitate large-scale adoption.

- **Multilingual Support:**

Adding multilingual capabilities would increase accessibility for users across different regions.

The App-Based Audiometer project is designed with sustainability in mind, ensuring its economic, environmental, social, and technical viability. Its cost-effectiveness, eco-friendly operations, accessibility, and scalability make it a future-ready solution for hearing loss detection and prediction. Furthermore, the project directly contributes to SDG 3: Good Health and Well-being by promoting early diagnosis, preventive healthcare, and accessible audiometry services, leading to improved public health outcomes. With continuous improvement and strategic partnerships, the project has the potential to positively impact global healthcare by making audiometry more affordable and accessible.

Chapter 6

Software Testing

The App-Based Digital Audiometer underwent a comprehensive software testing process to ensure its correctness, reliability, and user safety. Software testing was essential in validating both individual components and the integration of modules to guarantee a seamless user experience and accurate hearing assessments.

6.1 Type of Testing

The software testing for the App-Based Digital Audiometer involved two key types of testing: Unit Testing, Integration Testing and Acceptance Testing ensuring the reliability and correctness of the application.

- **Unit Testing**

Unit testing was a fundamental step in ensuring the reliability and accuracy of the App-Based Digital Audiometer. It involved verifying the functionality of individual software components in isolation, prior to their integration into the complete system. The goal of this testing phase was not only to check whether each unit or module of the codebase produced the expected outcomes but also to identify and resolve any logic errors or inconsistencies at an early stage of development. This helped in building a robust foundation for further integration and system-wide testing.

Each module was tested independently with predefined inputs to assess whether it returned correct and consistent outputs. Among the critical modules tested was the Tone Module, which is responsible for generating audio tones at precise frequencies (ranging from 250 Hz to 8000 Hz) and calibrated volume levels required for audiometric testing. This module played a vital role in simulating standard pure-tone audiometry conditions.

The Signal Processing Module was also tested to ensure it accurately measured and responded to background noise levels. This module included algorithms for real-time noise calibration and filtering, essential for maintaining the integrity of test results, especially in non-clinical environments with potential sound disturbances. Its functionality was validated to confirm that noise exceeding a certain decibel threshold (e.g., 40dB) triggered appropriate alerts and calibration mechanisms.

Another key component was the Audiometry Test Module, which managed the interactive flow of the hearing test. This module handled the logic for tone presentation and user response capture. Unit testing for this module focused on verifying whether user inputs (e.g., tapping a response button upon hearing a tone) were properly recorded and whether test progression logic (such as moving to the next frequency) was executed correctly.

Lastly, the Result Analysis Module was tested to ensure it accurately interpreted user responses and generated audiograms—visual graphs representing hearing thresholds

across frequencies for each ear. The module's calculations were validated to check for proper plotting, scaling, and interpretation of data, ensuring that audiograms accurately reflected the user's hearing ability.

Overall, unit testing provided an early assurance of quality by allowing developers to detect and resolve issues within each module before integration. It significantly reduced debugging time during later stages and ensured that each building block of the application met its intended functionality with precision.

- **Integration Testing**

Following unit testing, integration testing was carried out to verify that all individually tested modules worked together as a cohesive system. The objective of this phase was to ensure seamless data exchange and communication between the frontend user interface, the backend database (Firestore), and the signal processing functionalities. Integration testing played a critical role in identifying issues that could arise when modules interact, such as misaligned data formats, communication errors, or timing mismatches between asynchronous components.

Manual testing techniques were primarily employed during this phase, focusing on real-world use cases and functional interactions. For instance, the user registration flow was tested thoroughly to confirm that user credentials entered through the frontend were correctly authenticated and securely stored in the Firestore backend. This validated both the interface logic and the connectivity with cloud services.

One major area of focus was the audiogram data storage pipeline, which was tested to ensure that hearing test results were not only saved accurately in the database but could also be retrieved seamlessly for future analysis or visualization. Integration tests verified that test data maintained its integrity during transmission and that the data retrieval functions displayed correct information in the app interface.

The noise calibration and detection system was another critical integration point, where the signal processing module needed to work in harmony with the frontend UI to reflect real-time environmental noise levels. The system was tested to verify that it successfully adjusted calibration parameters based on noise input (e.g., showing warnings if ambient sound exceeded 40 dB), thereby maintaining the reliability of the hearing tests.

Additionally, headset detection was tested across multiple scenarios to confirm the accurate identification of both wired and Bluetooth headsets. The application was expected to recognize headset connections and adjust the calibration accordingly to account for hardware variations, ensuring consistent audio delivery. These integrations collectively validated that the app's components were not only functional on their own but also communicated effectively with each other when deployed as a unified system.

- **Acceptance Testing**

Acceptance Testing was carried out to ensure that the App-Based Digital Audiometer met both the functional expectations of the end users and the broader business objectives of the solution. This phase was crucial in validating that the entire system operated correctly in

real-world conditions and was ready for deployment in clinical or public health settings. The core objective was to confirm that the application behaved as expected from the user's perspective, providing a smooth and reliable experience throughout its key functionalities.

The testing process aimed to verify that all critical user scenarios—such as initiating a hearing test, responding to tones, viewing audiogram results, and accessing previously saved reports—could be completed successfully without technical difficulties. This ensured that the system covered essential use cases and that no major functional gaps existed. Additionally, acceptance testing served as the final checkpoint before the official release of the application, helping stakeholders make informed decisions regarding its deployment.

To achieve this, the testing was conducted by actual end-users, including audiologists and members of the target user group, such as patients with varying levels of hearing ability. The use cases were designed to mimic real-world situations, allowing testers to navigate the app as they would in practical scenarios. For example, testers followed workflows that involved launching the app, undergoing a hearing assessment, and viewing their results in the form of audiograms.

After the testing sessions, structured feedback was gathered through surveys and interviews. This feedback provided insights into user satisfaction, ease of navigation, and the clarity of instructions and outputs. The positive responses and successful completion of all user-centered tasks led to stakeholder approval, confirming that the application was ready for deployment and capable of delivering value in its intended environment.

6.2 Test cases & Test Results:

The app underwent rigorous testing, with detailed test cases covering functional and non-functional requirements. The Requirements Traceability Matrix (RTM) ensured that all requirements were mapped to specific test cases.

Table 6.1 Unit Testing Test Cases and Results Table

Test Case ID	Test Objective	Input	Expected Output	Actual Result	Status
TC-001	Verify user registration and login	User credentials	Successful sign-up/sign-in	Successful login	Passed
TC-002	Check background noise level detection	Background noise level (45dB)	Warning if noise > 40dB	Warning displayed	Passed
TC-003	Verify headset connection detection	Connected headset	Correct headset type identified	Headset detected properly	Passed
TC-004	Test headset calibration for volume adjustment	Initial volume settings	Correct volume calibration	Volume adjusted properly	Passed
TC-005	Conduct tone testing (250Hz to 8kHz)	Test tones	Accurate tone generation	Tones generated correctly	Passed
TC-006	Verify user feedback on tones heard	User feedback	Accurate result processing	Results processed correctly	Passed
TC-007	Verify audiogram generation	Hearing test data	Audiogram graph generated	Audiogram displayed	Passed
TC-008	Test result storage in Firebase	Test results data	Data stored securely	Data saved successfully	Passed
TC-009	Validate data retrieval from Firebase	Test ID	Accurate retrieval of stored results	Data retrieved correctly	Passed

Table 6.2 Integration Test Cases and Results Table

Test Case ID	Test Objective	Modules Involved	Input	Expected Output	Actual Result	Status
TC-001	Verify user registration with Firebase	User authentication & Firebase	User credentials	Successful sign-up/sign-in & data saved	Data stored successfully	Passed
TC-002	Validate noise detection and calibration integration	Noise detection & Calibration	Background noise level (45dB)	Calibration adjusted based on noise	Calibration successful	Passed
TC-003	Check headset detection and calibration integration	Headset connection & Calibration	Connecte headset	Volume calibration for detected headset	Calibration adjusted	Passed
TC-004	Validate tone generation and response tracking	Tone generation & Feedback module	Test tones (250Hz to 8kHz)	Accurate tracking of user responses	User responses tracked	Passed
TC-005	Verify audiogram generation from integrated results	Test results & Audiogram generation	Hearing test data	Audiogram graph generated and stored	Audiogram saved successfully	Passed
TC-006	Test data storage and retrieval from Firebase	Firebase & Test results module	Test result data	Results stored and retrieved correctly	Data stored and retrieved	Passed

Table 6.3 Acceptance Test Cases and Results Table

Test Case ID	Test Scenario	User Role	Expected Outcome	Actual Outcome	Status
AT-001	Complete hearing test flow	Patient	Test completed successfully with audiogram shown	Flow executed smoothly	Passed
AT-002	Admin views all users and test records	Admin	Admin dashboard shows all stored test results	Data displayed correctly	Passed
AT-003	Detect noise level and prevent test in noisy area	Patient	App shows warning and restricts test	Warning displayed correctly	Passed
AT-004	Verify headset detection and calibration adjustment	Patient	Correct calibration according to headset type	Calibration accurate	Passed
AT-005	Save and retrieve test results from Firebase	Admin/Patient	Results stored and accessed seamlessly	Retrieval successful	Passed

The App-Based Digital Audiometer underwent thorough software testing, covering unit testing, integration testing and acceptance testing. The tests verified that all modules functioned correctly, both individually and when combined. The successful execution of test cases ensured that the app met all functional and non-functional requirements, offering a reliable, accurate, and scalable solution for hearing assessments.

Chapter 7

Results and Discussion

7.1 Outcomes

The proposed app-based digital audiometer generates Audiograms. The data of 25 users collected from our app was compared with PTA and sensitivity and specificity was calculated based on these 25 users only. After validating the data of 25 users, the dataset of 1000 users was effectively processed, and 4 different classifiers RFC , SVM , XGBOOST and Naive Bayes were used to achieve high accuracy in classifying hearing loss severity for both ears. Evaluation metrics indicated reliable performance, with precision and recall values confirming the model's robustness. For the developed Application the confusion matrices for both the left and right ear tests provide insight into the classification performance of the proposed model.

Left Ear: The confusion matrix for the left ear (Fig. 3) demonstrates that the model correctly classified 9 normal cases and 13 hearing loss cases. It misclassified 1 normal case as hearing loss (false positive) and 1 hearing loss case as normal (false negative). The model achieved a high classification accuracy with a sensitivity of 93% and a specificity of 100%, indicating strong performance in detecting hearing loss while minimizing false positives.

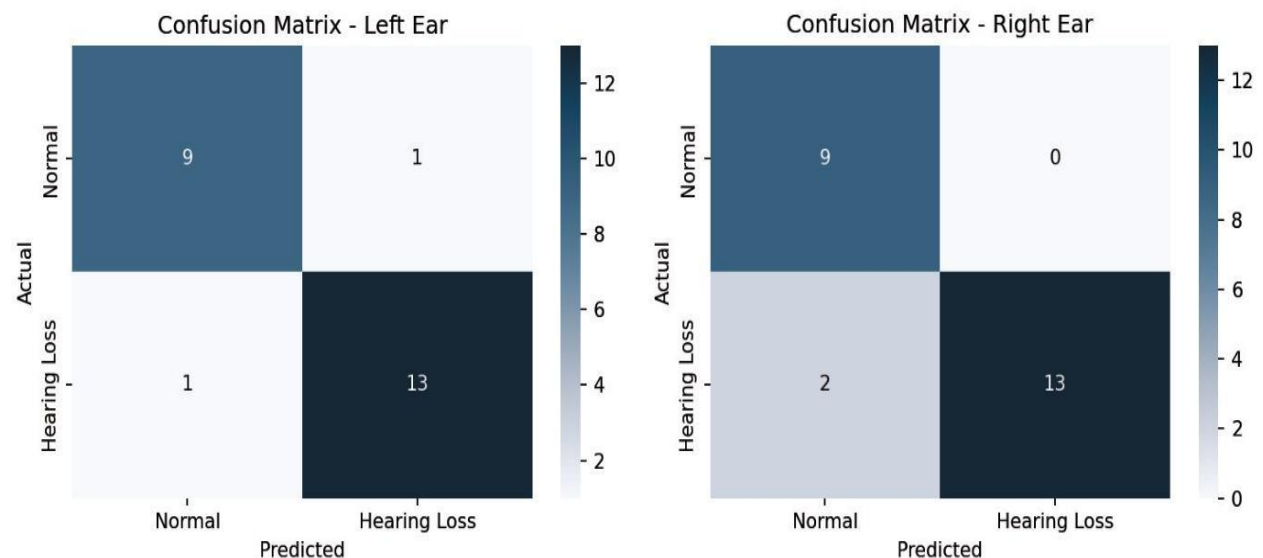


Fig 7.1 Confusion matrix of Left Ear and Right Ear

Right Ear: The confusion matrix for the right ear (Fig. 3) shows that the model correctly classified 9 normal cases and 13 hearing loss cases. However, it misclassified 2 hearing loss cases as normal (false negatives) while making no false positive errors. The right ear test resulted in a sensitivity

of 87% and a specificity of 100%, showcasing high reliability in distinguishing between normal hearing and hearing loss.

Performance of four different machine learning models—Random Forest(RFC), Support Vector Machine (SVM), XGBoost, and Naïve Bayes—for classifying audiometric data corresponding to both the left and right ears. The models were trained and tested separately on datasets for each ear, and their accuracy was recorded. Figure (learning curve-left ear) shows the learning curve of left ear and figure(learning curve-right ear) shows the learning curve of right ear for the 4 algorithm (RF, SVM, XGBoost, Naive Bayes). The results of our experiments are presented in (Table comparison).

7.2 Result analysis and validations

The evaluation results indicate that SVM consistently outperforms other models, achieving the highest accuracy across both left and right ear datasets. On the left ear dataset, SVM attained a training accuracy of 96% and a test accuracy of 96%, whereas on the right ear dataset, the model improved slightly, reaching a training accuracy of 97% and a test accuracy of 96%. This suggests that SVM is well-suited for audiometric classification, likely due to its ability to effectively handle high-dimensional feature spaces.

Table 7.1 Model Accuracy on Left and Right Ear

Model	Left Ear Train Accuracy	Left Ear Test Accuracy	Right Ear Train Accuracy	Right Ear Test Accuracy
Random Forest	0.95	0.94	0.94	0.93
SVM	0.96	0.96	0.97	0.96
XGBoost	0.94	0.93	0.95	0.94
Naive Bayes	0.89	0.89	0.87	0.87

The XGBoost model also performed well, achieving a training accuracy of 94% and a test accuracy of 93% for the left ear, while on the right ear, the accuracy increased to 95% and 94%, respectively. This indicates that XGBoost generalizes well and maintains stable performance across different datasets, making it a reliable model for classification tasks involving hearing data.

The Random Forest model demonstrated comparable accuracy, with a training accuracy of 95% and a test accuracy of 94% for the left ear, while its performance slightly declined on the right ear, where it achieved 94% and 93% accuracy for training and testing, respectively. Although Random Forest showed strong predictive ability, its slight drop in test accuracy for the right ear suggests it may be more prone to minor overfitting compared to SVM and XGBoost.

The Naïve Bayes model exhibited the lowest accuracy across both datasets, with an accuracy of 89% for both training and testing on the left ear and a slight decrease to 87% for the right ear. This result aligns with expectations, as Naïve Bayes operates under the assumption of feature independence, which may not hold true for audiometric data, potentially limiting its classification performance.

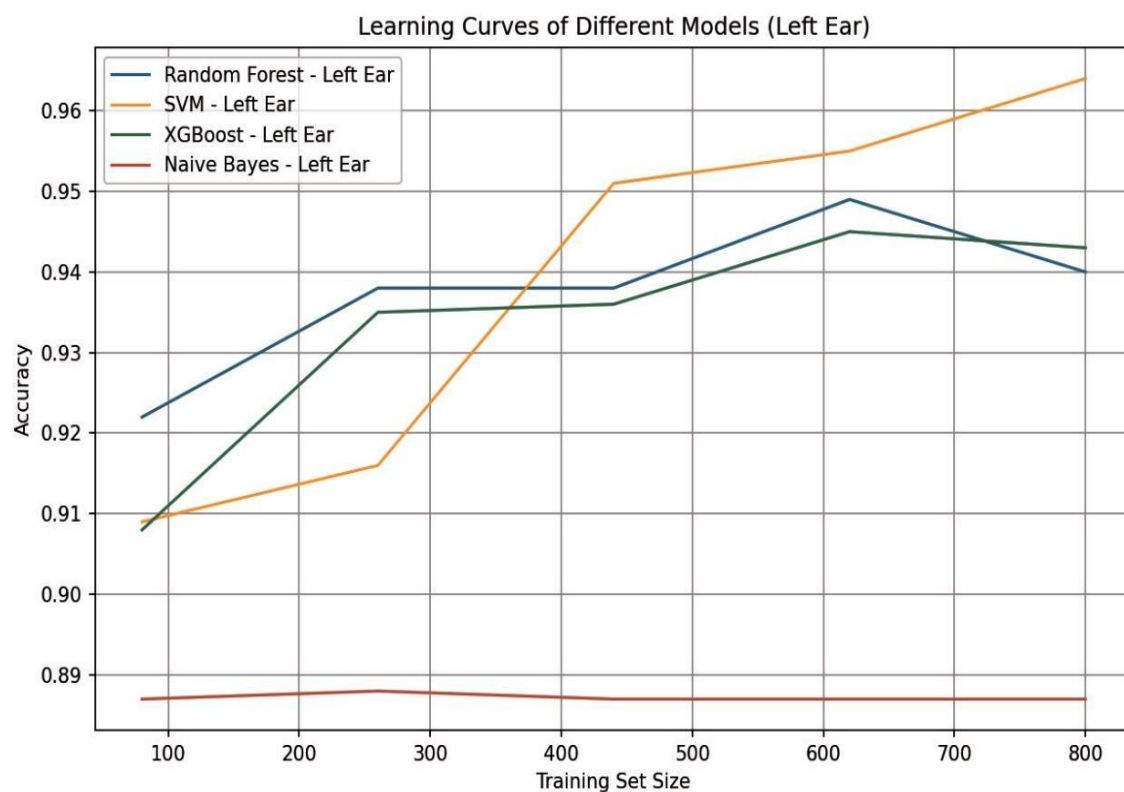


Fig 7.2 Learning Curve (Left Ear)

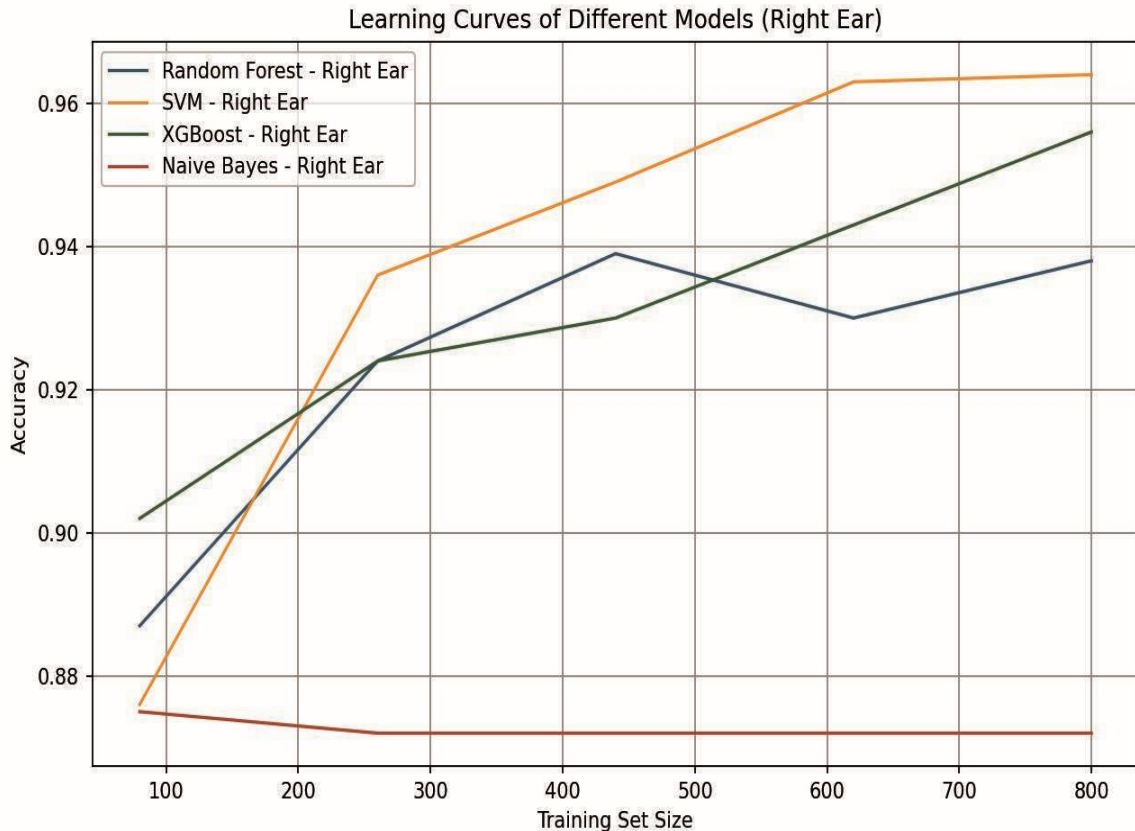


Fig 7.3 Learning Curve (Right Ear)

The learning curves of the models provide deeper insights into their generalization behavior as the training set size increases. From the plotted learning curves (Figure X), we observe the following trends:

- SVM:** shows a steep increase in accuracy with additional training data, reaching saturation around 600-700 samples, where it achieves the highest accuracy. This suggests that SVM benefits significantly from larger datasets, making it ideal for audiometric classification tasks with sufficient training data.
- XGBoost:** Demonstrates a steady rise in accuracy with increasing training samples, indicating its ability to generalize well across different dataset sizes. The model shows no signs of overfitting and maintains stable performance.
- Random Forest:** exhibits a slight fluctuation in accuracy beyond 500 samples, suggesting potential overfitting or sensitivity to dataset variations. Despite this, it remains a strong performer.
- Naive Bayes:** Remains largely flat with minimal improvement in accuracy, indicating that it does not benefit significantly from additional data. This suggests that its simplistic probabilistic assumptions may not be suitable for complex audiometric patterns.

These learning curve observations confirm that SVM and XGBoost not only achieve the highest

accuracy but also demonstrate the best scalability with increasing data sizes, making them the most robust models for the given classification task. A comparative analysis of left and right ear performances reveals minor variations in accuracy across models. Notably, SVM and XGBoost showed improved performance on the right ear, while Random Forest and Naïve Bayes exhibited a slight decline. The increase in accuracy for SVM and XGBoost suggests that the right ear dataset may have more distinguishable patterns that these models were able to capture effectively. Conversely, the minor drop in Random Forest's accuracy may indicate slight overfitting on the right ear dataset. These findings suggest that SVM and XGBoost are the most reliable models for audiometric classification. Future work could explore hybrid models or deep learning approaches to further enhance classification performance.

Table 3 presents a comparison of various audiometry studies, focusing on dataset usage and performance metrics. Study [4] utilized data from 70 participants, achieving classification accuracies of 97.4% (<35 dB), 22.2% (35–64 dB), and 38.5% (>65 dB) but did not incorporate ML and proper calibration of device used. Study [9] analyzed 256 audiograms, reporting sensitivity of 93.8% and specificity of 82.2%. They used external devices for calibration. But in our proposed system the calibration can be carried out in built from mobile device. Study [10] examined 250 patients, achieving a sensitivity of 29% and specificity of 100%, without ML integration. But our proposed solution showed more accuracy. Study [11] developed an ML-based hearing loss prediction model using its own dataset, integrating an ML model within a user interface and achieving test and train accuracies of 92.6% and 93.625%, respectively. Study [20] used data from 1644 people, RFC, Adaboost, MLP, SVM achieved an overall accuracy of 79.7%, 78.6%, 78.9%, 80.1% respectively indicating a reliable performance in predicting hearing impairment, without application-level performance details. The proposed model, based on 1000 audiograms, demonstrates left ear testing accuracy of RF: 94%, SVM: 96%, XGBoost: 93%, and Naïve Bayes: 89%, while the right ear testing accuracy is RF: 93%, SVM: 96%, XGBoost: 94%, and Naïve Bayes: 87%. This analysis highlights the importance of dataset quality, sensitivity, specificity, and ML integration in advancing digital audiometry, showcasing the proposed model's potential for enhanced hearing loss detection.

Table 7.2 Performance Table

Ref.	Dataset	Performance	
		Application	ML
[4]	70 participants	Class A(<35dB):97.4% Class B(35-64dB):22.2% Class C(>65dB):38.5%	NA
[9]	256 audiograms	Sensitivity: 93.8% Specificity:82.2%	NA
[10]	250 Patients	Sensitivity 29% Specificity 100%	NA
[11]	Own dataset	User Interface integrated with ML model	Test accuracy: 92.6% Train accuracy:93.625%
[20]	1,644 patients	NA	RF:-79.7% Adaboost:-78.6% MLP:-78.9% SVM:-80.1%
Proposed Model	1000 Audiograms	Left Ear:- Sensitivity: 93% Specificity:90% Right Ear:- Sensitivity: 87% Specificity:100%	Left Ear: RF: 94%, SVM: 96%, XGBoost: 93%, Naive Bayes: 89% Right Ear: RF: 93%, SVM: 96%, XGBoost: 94%, Naive Bayes: 87%

7.3 User Interface

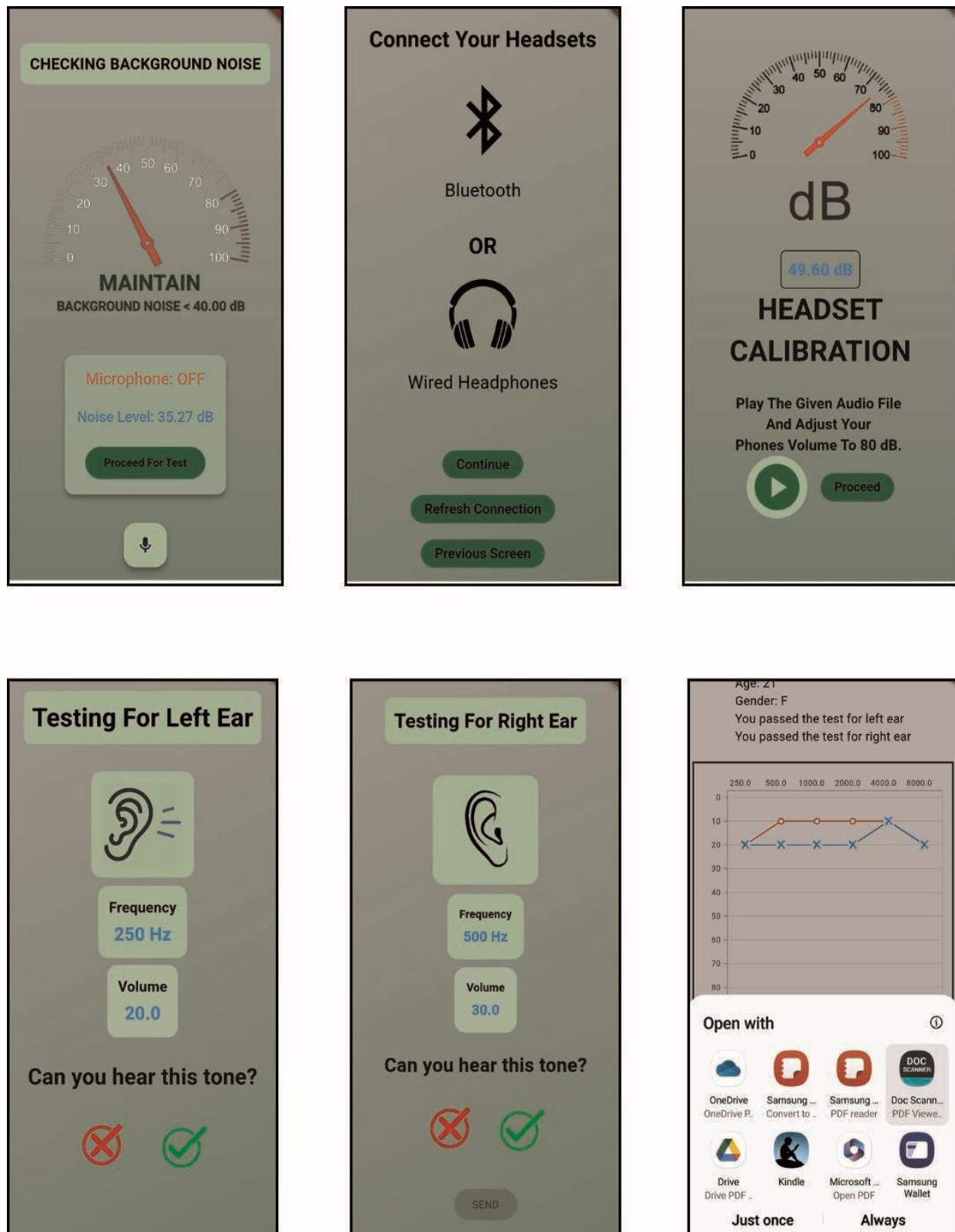


Fig 7.4 Individual Testing

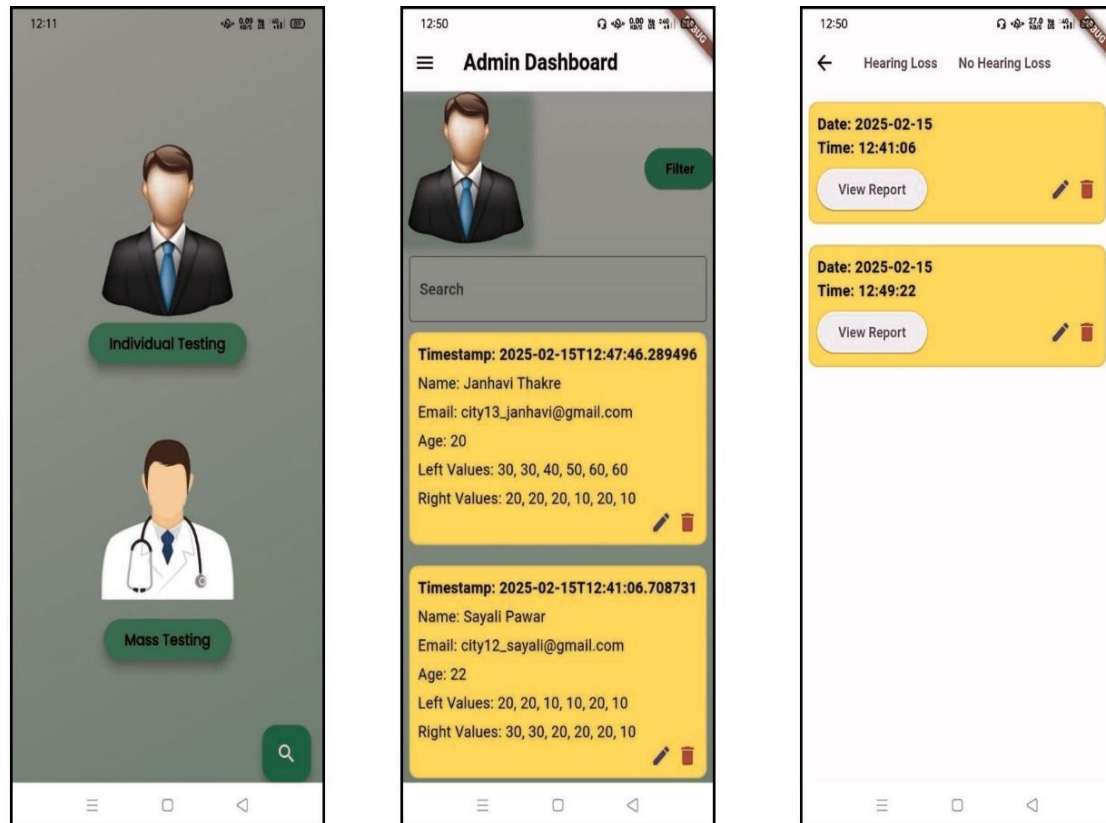


Fig 7.5 Mass Testing

Chapter 8

Conclusion & Future Scope

8.1 Conclusion

The successful completion of this project marks an important step forward in the field of digital healthcare, specifically in the domain of hearing assessment and early diagnosis. By combining mobile technology with machine learning, this project introduces two practical and innovative solutions: an App-Based Digital Audiometer and a Machine Learning (ML) model for predicting hearing loss. Both solutions work in tandem to address key limitations of traditional audiometry systems and aim to make hearing care more accessible, accurate, and scalable.

Hearing loss is a growing health concern worldwide, and early detection is crucial for effective treatment and intervention. However, conventional hearing assessment methods are often costly, require sophisticated equipment, and must be conducted by trained professionals in clinical environments. These constraints make it difficult for people in rural, remote, or economically challenged regions to receive timely care. The App-Based Digital Audiometer developed in this project directly tackles these issues. By transforming a regular smartphone into a portable audiometric testing tool, it eliminates the need for bulky equipment and specialized setups. This makes hearing tests available at the fingertips of users, reducing dependency on physical infrastructure.

The application is designed with a user-friendly interface, allowing individuals of all ages to perform basic hearing tests independently. It also supports bulk screenings in community centers, schools, and health camps, enabling large-scale outreach programs without significant logistical effort. This democratization of hearing assessment ensures that help reaches the people who need it most, especially those who otherwise wouldn't have access to audiological services.

A significant contribution of this project lies in its use of real-world data. Through the app, hearing test data from over 1,000 users was collected and analyzed. This dataset served as the foundation for developing an intelligent ML model capable of predicting hearing loss severity. Several widely-used algorithms, including Random Forest Classifier (RFC), Naive Bayes, Support Vector Machine (SVM), and XGBoost, were trained and evaluated. These models analyzed a combination of audiometric thresholds and demographic features to determine the extent of hearing impairment in users.

The predictive model demonstrated high accuracy, which highlights its potential as a decision-support tool in hearing healthcare. Instead of relying solely on raw hearing test results, users can now receive a detailed and data-driven prediction about their hearing condition. This added intelligence empowers users to seek medical attention at the right time and allows healthcare providers to prioritize patients based on predicted severity.

More broadly, this project reflects how emerging technologies can be used to address persistent public health challenges. The combination of mobile app development and machine learning is not just technically sound but also socially relevant. It sets the stage for future innovations in

remote diagnostics and AI-based health monitoring. As more healthcare services shift towards digital platforms, such solutions can play a critical role in supporting early diagnosis, preventive care, and health awareness.

From a research perspective, this project opens up multiple avenues for future work. Features like real-time doctor consultations, cloud-based report storage, integration with Bluetooth hearing aids, and multi-language support can enhance the app's usability and reach. On the ML side, the model can be continuously trained with new data to improve accuracy and adapt to diverse populations. This would also enable the system to detect more specific patterns and rare hearing conditions over time.

In conclusion, the project successfully addresses the dual challenges of accessibility and intelligence in hearing care. It proves that with the right blend of technology, innovation, and user-centric design, healthcare tools can be made affordable and impactful. The App-Based Digital Audiometer and ML-powered predictive model together provide a complete, practical, and scalable solution for modern audiological assessments. This work not only fulfills the academic and technical goals of the project but also contributes meaningfully to society by laying the foundation for more inclusive, data-driven, and tech-enabled healthcare solutions.

8.2 Future Work

Future enhancements to the platform will focus on expanding the dataset, optimizing ML algorithms for greater accuracy, and expanding its capabilities and accessibility. The planned integration of a sophisticated chatbot powered by natural language processing will provide real-time user support, addressing common questions and technical issues immediately. This AI-driven assistance will help reduce user anxiety and improve test completion rates. Additionally, in future the development of multilingual options will break down language barriers, making the platform accessible to diverse populations globally.

8.3 Applications

Healthcare & Telemedicine – Remote and clinical hearing assessments.

Public Screening – Mass testing in schools, workplaces, and rural areas.

Occupational Safety – Hearing monitoring in noisy industries.

Personal Use – Self-assessment and hearing health tracking.

Research & AI – Data-driven audiology advancements.

APPENDIX

Appendix A: Details of Paper application

Paper	Title	Conference/Journal	Status
1	App-Based Audiometer for Hearing Loss Detection and Prediction Using Machine Learning	9th International Conference on Control Communication, Computing and Automation	Submitted

Paper 1: Proof of submission

9th International Conference on Control Communication, Computing and Automation : Submission (477) has been edited.

1 message

Microsoft CMT <noreply@msr-cmt.org>
To: gauravmahale.1011@gmail.com

Wed, Apr 16, 2025 at 1:48 PM

Hello,

The following submission has been edited.

Track Name: Cognitive Computing and Machine Learning

Paper ID: 477

Paper Title: App-Based Audiometer for Hearing Loss Detection and Prediction Using Machine Learning

Abstract:

This research delves into the domain of health and medical diagnosis, aiming to enhance accuracy and efficiency in diagnostic processes. This study looks into the development of an app and prediction of hearing loss using machine learning (ML) approaches based on age and audiometric data. The app collects data and passes it to the ML model for analysis and comparison of left and right ear results. The primary issue is that conventional physical audiometers are both costly and time intensive. Moreover, the expense of establishing a pure tone is excessively high. The proposed solution involves a digital audiometer integrated into an app, designed to overcome challenges, enable mass testing, and offer portability. It collects data, which is then utilized for predictions using the Random Forest algorithm (RFC), Naive Bayes, SVM and XGBoost. The results demonstrated high accuracy making it easy for early detection and intervention.

Created on: Wed, 16 Apr 2025 08:05:30 GMT

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Secondary Subject Areas: Not Entered

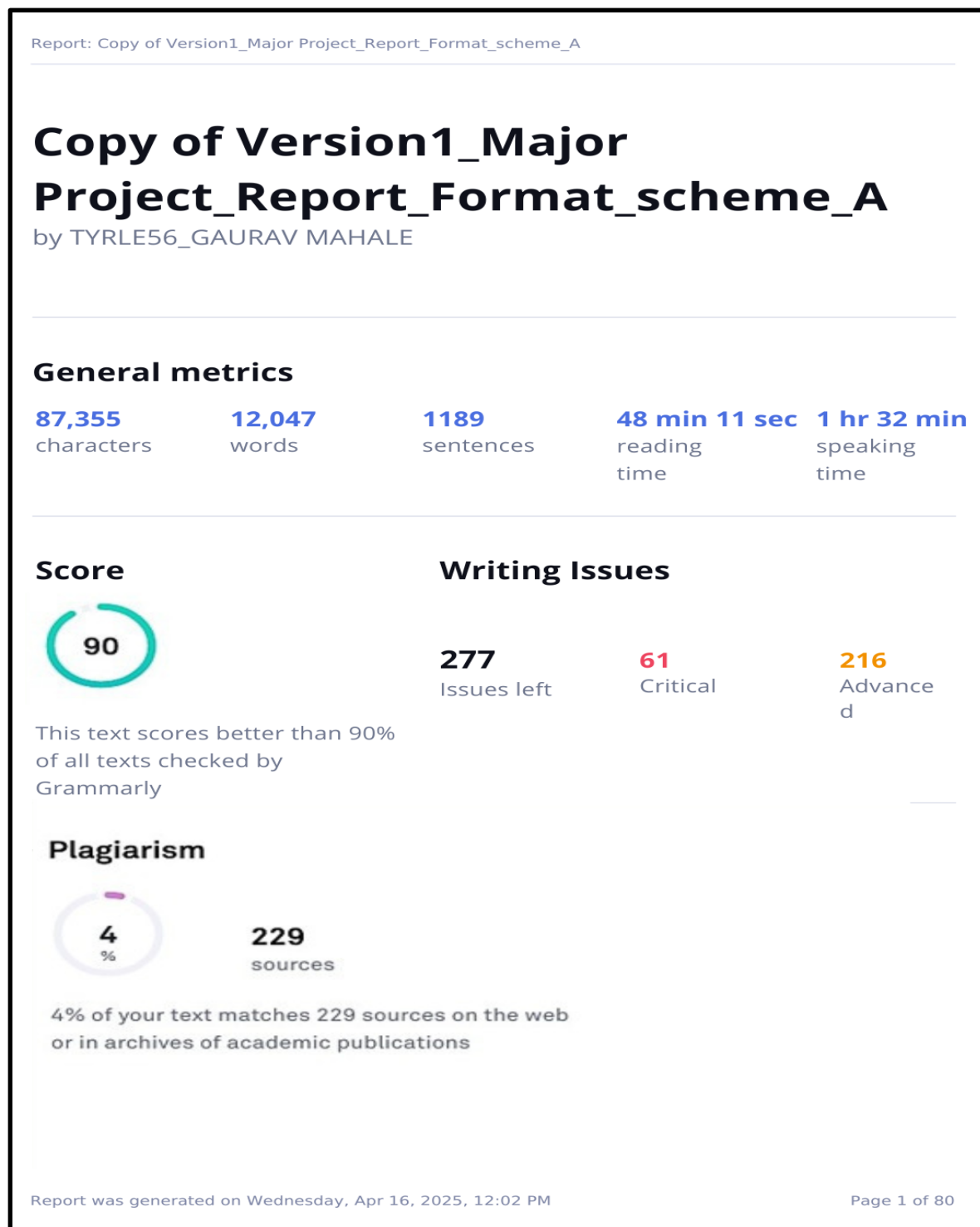
Submission Files:

App-Based Audiometer for Hearing Loss Detection and Prediction Using Machine Learning_compressed.pdf (343 Kb, Wed, 16 Apr 2025 08:04:57 GMT)

Submission Questions Response: Not Entered

Thanks,
CMT team.

Appendix B: Plagiarism report of the project report



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